THE FORBIDDEN PLANTS

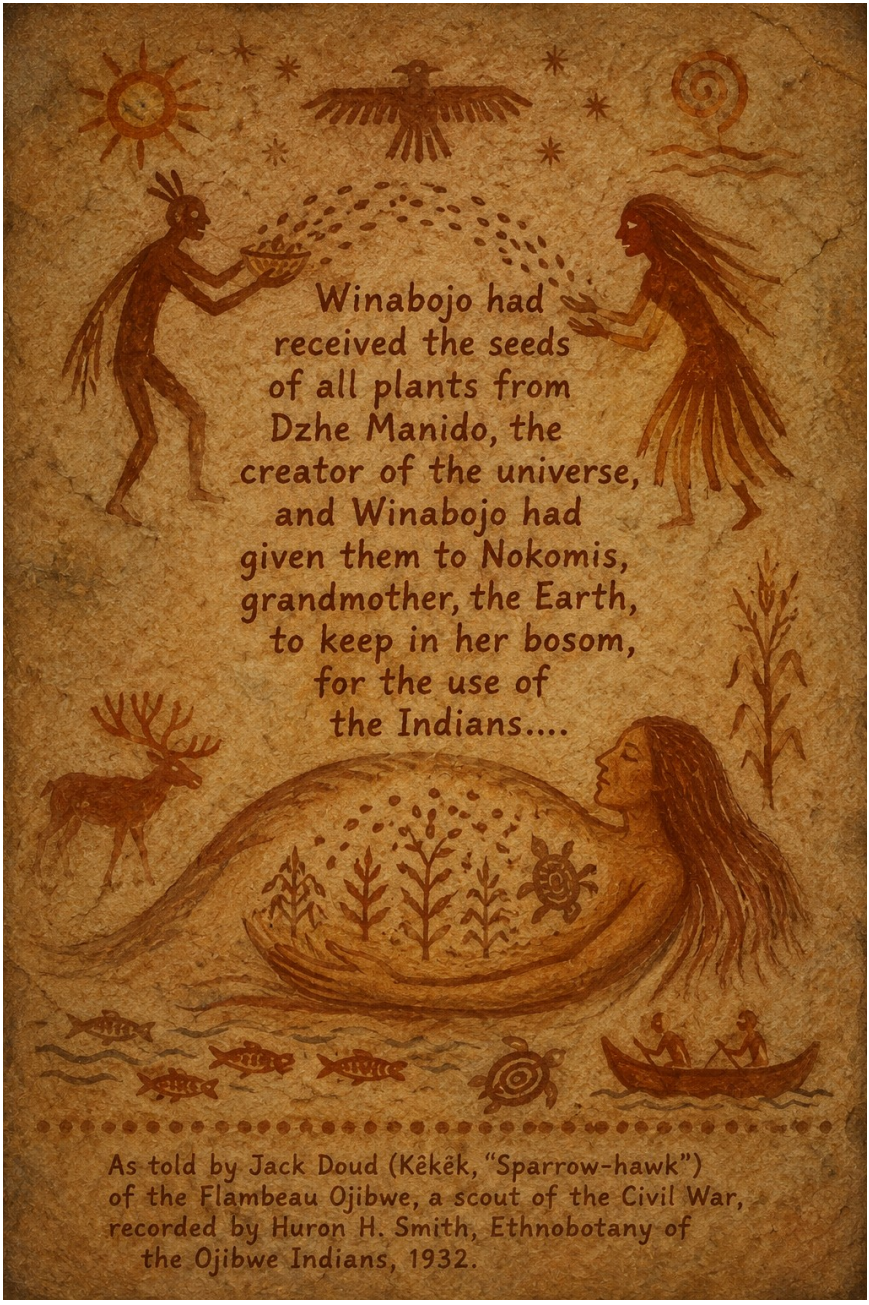
Outlawing the Sacred and Selling the Deadly: How a Nation's Plants Were Criminalized by Money and Power

Book Three of The Forgotten Series

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For all the narrative dissidents who have done their homework, and who have come to realize that the difference between conspiracy and reality is about six months.



As told by Jack Doud (Kêkêk, "Sparrow-hawk")
of the Flambeau Ojibwe, a scout of the Civil War,
recorded by Huron H. Smith, Ethnobotany of
the Ojibwe Indians, 1932.

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Introduction: From a World That Worked

Begin with a people. The Anishinaabe of the Great Lakes lived on a generous and evolving land, and they lived on it well. More than a hundred plant species gave them their food, their medicine, their dye, and their craft, and a self-sufficient life was built on that abundance.

What the summer poured forth from the gardens, the Odawa Andrew Blackbird remembered from his boyhood as sheer abundance. The Arbor Croche country, he wrote, was mostly grassy plain, and it carried "such an abundance of wild strawberries, raspberries and blackberries that they fairly perfumed the air of the whole coast with fragrant scent of ripe fruit. The wild pigeons and every variety of feathered songsters filled all the groves, warbling their songs joyfully and feasting upon these wild fruits of nature." "I never knew my people to want for anything to eat or to wear," he wrote of those summers, "as we always had plenty of wild meat and plenty of fish, corn, vegetables, and wild fruits."

Long before written words, the peoples of the Great Lakes kept a body of knowledge as exact as any science, held in their hands, their memories, and the materials the land provided: the wisdom of adaptation, of attention, of living in a harmonized relationship with their environment, and of carrying that knowledge from one season into the next. They did not sentimentalize the plants. They ate them, burned them, brewed them, dosed themselves with them, and buried them with their dead. But they used them inside a bargain, take what you need and return what you owe, and the bargain held for thousands of years.

That world, the plants and the people who knew them, is the subject of the two volumes that come before this one in The Forgotten Series. What follows here stands on its own, and asks nothing of the reader that the others gave. It is about what was done to that bargain, and to the plants that lived, for better or worse, within a niche closely associated with humanity's needs.

The argument is plain, and it can be put in a single sentence: which plants a nation forbids and which it sells has rarely been decided by how dangerous they are. It has been decided by money, by race, by moral panic, and by power. The plants bound in the public mind to people without power were made into crimes, and the people who used them were punished for it. The plants that turned a profit for people with power were licensed, taxed, and advertised. This was true even when the profitable plant killed by the hundreds of thousands and the forbidden one, in its traditional form, killed almost no one. Tobacco and alcohol, the two deadliest drugs in the country, were legal. The coca leaf in an Andean cheek, which has never killed anyone that way in four thousand years of use, is contraband the nations of the world have sworn to abolish. The difference between the two fates was not danger. It was who profited and who paid.

The natural objection to an argument like this is that it sounds like a conspiracy theory, that corporations and governments do not really sit in a room and coordinate to bury the truth about their products and criminalize their rivals. The answer is to be determined by the reader, for what follows carries no opinions, just documented facts found within court records and government ledgers. What this account describes is a pattern, not a plan. The documents represent a transparent system whose workings speak through actions, not innuendo. It is the same method turning up again and again, in different hands, in different decades, against different plants, because the incentives that produced it were the same each time. No secret meeting is required to explain it, any more than a secret meeting is required to explain why water runs downhill. Where the record shows coordination, and in a few cases a federal court has found exactly that, the coordination is named and the finding is cited. Everywhere else the claim is only what the evidence will bear: a pattern too consistent, across too many countries and too many years, to be an accident or due to incompetence.

The second thing to be clear about is where the evidence comes from. This is not a book of suspicions. To be clear, the spine is the documented, public record: the findings of courts, the reports of governments, the studies in the medical literature, and above all the internal documents that lawsuits pried out of company files, in which industries wrote down in their own words what they knew and when they knew it. The citations are not decoration. They are the argument's armor, and the argument is built to stay strictly to the available facts. Where the record proves a thing, this account asserts it. Where the record proves only that two things sat next to each other, it says so, and calls that an adjacency rather than a scheme. The plants that were forbidden are never pretended to be harmless; several of them can kill, and the pages that follow say plainly how. The point is not that the forbidden plants are safe. The point is that safety was never what decided their fate.

One shape recurs often enough in these chapters to be given a name: the same interests profit when people are made sick, and profit again when the sick try to get well. An industry sells the addiction and then sells the cure for it; it engineers the illness into the food supply and then sells the treatment; it seeds an epidemic with one product and patents the remedy for another. Where that second profit is a proven scheme, this account calls it a scheme. Where it is only an industry standing conveniently beside a harm it helped create, it says so. The adjacency alone is damning enough, and it does not need to be inflated into a plot.

Almost every plant in these pages began as a gift. Tobacco was a sacrament laid in the ground before the hunt and the harvest. Corn was a staple and a sacred food. The coca leaf held an empire's runners on their feet across the Andes; peyote, the Amazonian vine, and the Mexican mushroom were roads to the sacred, kept by peoples who had no industry and no lobby to defend them; the opium poppy and cannabis were medicines older than any pharmacy. Each of them was turned, by money and power, into either a weapon or a crime, and usually into whichever one provided greater

profit. That inversion, a gift made into a poison or a felony, is the story these chapters tell, plant by plant.

It tells it in four parts. The first, Tobacco and Corn, opens not on a prohibited drug but on two sacred plants of that older world, and watches them corrupted into the deadliest legal products in the country: the plants the powerful were allowed to keep, because they paid. The second part, The Forbidden Plants, takes up the ones that were criminalized, cannabis and coca, peyote and ayahuasca and the mushroom and the poppy, and shows in each case that the ban tracked the people who used the plant rather than the harm the plant did. The third part, The Playbook, sets down the method itself, proven in the records of leaded gasoline, tobacco, and the forever chemicals, and then shows the same machine carried into the food on the family table. The fourth part, The Reckoning, gathers the receipts, names who paid, and follows the bill to the place it finally comes due.

That place is where the story began. The people who lived by returning what they took now carry the highest rates of diabetes, of alcohol death, and of drug overdose of any people on the continent. The cause is not in their blood; it is in the world that was built on top of theirs, an environment engineered at every level to profit from what it does to the human body. Their reciprocal world was not merely lost. It was replaced by its opposite, and the sickness of that replacement falls hardest on the very people it displaced. That is where these pages end. It begins with a single sacred plant, and with the men who learned how much money could be made by teaching a nation to smoke it.

PART ONE

Tobacco and Corn



Chapter One

Tobacco



Before any of what follows, before the tax stamps and the surgeon general and the executives who swore under oath that they did not believe the product was addictive, there was a plant that the peoples of this continent held sacred: a little tobacco laid in the ground before a root was dug, a pinch given to the water before a hard crossing, a leaf offered to an elder in return for a song or a piece of knowledge. It was not smoked for pleasure so much as offered for a purpose. It carried words to the spirits. Among the Anishinaabe it was the first of the sacred medicines, the thing a person put down before asking anything of the land, and the record recounts the rites: the tobacco set at the foot of the birch before the bark was taken, the tobacco spoken to a plant that was about to be made into medicine, the tobacco burned so that its smoke would carry words upward. A people who believed they owed the land reciprocity: for everything they took from it they paid a great part of that debt in tobacco.

The plant of the ceremony was often not the plant of the modern cigarette. The tobacco raised for the pipe and the offering was frequently the stronger, harsher native species the botanists call *Nicotiana rustica*, grown in small quantities and used with restraint. A man did not carry it to enjoy it. He carried it because it was the currency of a relationship, the thing that made a request to a greater power legitimate, the way a gift makes a request legitimate between people. It was smoked in council so that the smoke passing from mouth to mouth would bind the men who shared it to what they had agreed. It was offered across a fire to seal a peace, and the pipe that held it, in many nations, was the single most sacred object a community owned, handled only with ceremony and never for pleasure. To offer tobacco was to speak carefully. Nothing about that use resembled a pack a day.

And yet that plant and the plant in the cigarette are the same genus, and in the space of a few generations the people's ceremonial offering, demonstrating respect and gratitude, was subverted for profit. And much worse still, tobacco turned into the deadliest consumer product in the history of

the country. That remaking, and the machinery that accomplished it, is the thesis of this whole book, which is why we begin here, with the one plant in these pages that was never forbidden.

The arithmetic is stark. Cigarette smoking kills on the order of four hundred and eighty thousand Americans every year, a figure the Centers for Disease Control and Prevention has published and republished for a generation. That is more than alcohol, more than every illegal drug combined, more than traffic crashes and gun deaths and house fires together. It is, year after year, as if a full commercial airliner crashed and killed everyone aboard every few hours, forever. By any measure a public-health authority would recognize, tobacco is the deadliest consumer product in the history of the country. And it has never been banned. It was sold at every gas station and grocery counter in America through every year of the war on drugs, while young men went to prison for the possession of plants that had killed no one. Tobacco did not come under the authority of the Food and Drug Administration, the agency that governs what may be sold as a drug, until 2009, which was forty-five years after that same government first told the public, officially and in a report of its own, that the product caused cancer.

These facts do not sit together: the discrepancy between this harmful substance and the regulations imposed on it is one of non-equivalence. If the danger of a plant decided whether it was allowed, tobacco would have been the first plant forbidden. It was instead the last to be even lightly regulated, and it remains legal today. Every prohibited plant in the chapters that follow, the coca leaf a treaty ordered abolished, the peyote a church had to defend before the Supreme Court, the cannabis that filled the jails, will be measured against this one. The question we will keep asking is not, in the end, why those plants were banned. It is why this one was not. The answer, here and in every chapter after, is that this plant paid.

It was not named sacred by accident, and it did not become deadly by an uncertain course of events either. Both were clearly premeditated, but the second was also deliberate, facilitated with the help of the two institutions a person is most inclined to trust: the physician and the press.

For the first half of the twentieth century the cigarette advertised itself through the doctor. The historians Martha Gardner and Allan Brandt, studying physician imagery in cigarette advertising for the *American Journal of Public Health* in 2006, traced how the brands spent the years from the nineteen-thirties into the nineteen-fifties draping themselves in the white coat. Lucky Strike told the country that "20,679 Physicians say 'Luckies are less irritating.'" Camel introduced the "T-Zone," the throat and taste a doctor supposedly approved. Philip Morris and Chesterfield ran their own versions, each pressing the same suggestion: that a cigarette was a considered, even a healthful, choice, and that the men in white coats had explored it. No doctor had studied any such thing. The point was never the study. The point was the association, the quiet capitulation of medical authority to a marketing campaign: the white coat borrowed to sell the very product the profession would, in time, have to warn the country against. The institutions meant to stand between the public and a manipulated market were not yet in the room, for the drug that would come to kill more Americans than any other had, as yet, no regulator with the power to touch it. Medicine's good name was lent to a product medical authorities would eventually call a poison.



Lucky Strike advertisement, 1930. The physicians' endorsement was produced by handing out free cartons and asking the recipients which brand they smoked.

The purest example ran under R. J. Reynolds. In 1946 the company began advertising its Camel brand with a line it would ride for the better part of a decade: "More Doctors Smoke Camels Than Any Other Cigarette." The advertisements showed physicians in white coats, and they cited a survey, a real survey with a real and impressive number, one hundred and thirteen thousand five hundred and ninety-seven doctors. What the advertisements did not

explain was how the survey had been taken. Reynolds had retained an advertising firm to hand out free cartons of Camels to physicians at medical conventions, and then to ask those same physicians, the free cigarettes already in their hands, which brand they smoked. The number was genuine. The impression it was built to create, that the medical profession had studied cigarettes and endorsed this one, was manufactured. A version of this trick appears in every chapter that follows: not a lie exactly, but a true fact arranged to produce a false belief.

The campaign worked because it borrowed authority the industry did not have and lent it to a product the industry already had reason to doubt. For most of the century the doctor in the advertisement and the science in the laboratory pointed in opposite directions, and the companies knew it.

The science was not, in fact, in much doubt, and it had not been for some time. In 1950 two pairs of researchers, working independently and an ocean apart, reached the same conclusion by different roads. In the United States, Ernst Wynder, then a medical student, and the thoracic surgeon Evarts Graham interviewed hundreds of lung-cancer patients about their habits and found almost all of them to be heavy, long-term smokers; they published the finding in the Journal of the American Medical Association. Graham himself was a smoker, and doubted his own result at first; he later quit, and later still died of the disease he had helped to prove. In Britain that same year, Richard Doll and the statistician Austin Bradford Hill published a parallel study of London hospital patients. Doll and Hill then did the more durable long-term study: beginning in 1951 they enrolled some forty thousand British doctors and followed them for years, recording who smoked and who did not and watching who died, and the smokers died of lung cancer at rates that no amount of chance could explain. By the middle of the nineteen-fifties the link between cigarettes and lung cancer was firmly established. Predictions generated from the hypotheses confirmed the negative effects. The industry's

response was not to change the product. It was to change the public.

On the fourth of January, 1954, the major tobacco companies ran a single advertisement, jointly, in hundreds of newspapers across the country, reaching an estimated forty-three million Americans. It was called "A Frank Statement to Cigarette Smokers," and it had been shaped by John Hill of the public-relations firm Hill and Knowlton. Its tone was one of grave and fatherly concern. "We accept an interest in people's health as a basic responsibility, paramount to every other consideration in our business," it announced, and it promised that the industry itself would fund honest, disinterested research to settle the question. To that end it created a research body, the Tobacco Industry Research Committee, later renamed the Council for Tobacco Research, which would look into the matter properly.

A close reading of the Frank Statement illustrates the method, because it is the founding document of a technique recurrent in every chapter after. It is not a lie. Every sentence in it is either true or unfalsifiable. It does not claim that cigarettes are safe; it claims that the question is still open, and it positions the industry as the responsible party working to answer it. The research body it promised did not exist to answer the question. It existed to keep it open: to fund the studies that pointed elsewhere, to underwrite the scientists who would testify to uncertainty, to publicize every stray finding that muddied the picture, and to ensure that for as long as possible a reasonable, careful person could believe the jury was still out. The historian Robert Proctor, who spent years inside the industry's own archives for his book *Golden Holocaust*, gave this practice a name: agnotology, the deliberate manufacture of ignorance. It is the exact inversion of how we imagine corporate lying works. The industry did not have to prove its product safe. It had only to make the danger look uncertain, and uncertainty, it turns out, can be manufactured for decades and sold to the public as balance and fair play.

It worked for ten years. When the United States government finally spoke, it did so carefully. On a Saturday, the eleventh of January, 1964, chosen deliberately to fall on a day when the stock markets were closed and the weekend papers thin, Surgeon General Luther Terry released *Smoking and Health*, the report of an advisory committee of ten scientists who had reviewed more than seven thousand studies. Its central conclusion was blunt: cigarette smoking causes lung cancer, and the matter was one for "appropriate remedial action." The market, when it opened, barely moved. The manufacture of doubt had done its work so well that the public had been trained to treat a settled fact as a living controversy, and a single Saturday report changed few habits. Sales dipped and recovered within the year. Over the next several years Congress required a warning label on the side of the pack and, in a law that took effect at the beginning of 1971, ordered cigarette advertising off television and radio. The industry absorbed the loss of the airwaves as a cost of doing business, shifted its spending to print and sponsorship, and kept selling.

A FRANK STATEMENT TO CIGARETTE SMOKERS

Recent reports on experiments with mice have given wide publicity to a theory that cigarette smoking is in some way linked with lung cancer in human beings.

Although conducted by doctors of professional standing, these experiments are not regarded as conclusive in the field of cancer research.

We believe the products we make are not injurious to health.

We always have and always will cooperate closely with those whose task it is to safeguard the public health.

Signed by the presidents of the leading cigarette manufacturers, published in hundreds of newspapers, January 4, 1954.

What no one outside the industry could yet prove was that the companies had known the truth the entire time, and had chosen to say the opposite. That proof, when it came, came on paper records, and it came because of one man's conscience and considerable nerve. Merrell Williams was a

paralegal at a Louisville law firm, Wyatt, Tarrant and Combs, that did work for Brown and Williamson. His job, for several years beginning in the late nineteen-eighties, was to sort and code the company's internal documents for litigation, and as he read them he understood what he was holding. He began copying them, thousands of pages, and carrying the copies out of the building. It was a serious risk, professionally and legally, and it cost him; he later suffered a heart attack he attributed partly to the strain. Eventually the copies reached Stanton Glantz, a scientist at the University of California, San Francisco. They arrived in a box from an anonymous sender who signed "Mr. Butts", after the cartoon cigarette. Glantz and his colleagues authenticated and published them as *The Cigarette Papers*.

What the documents showed was that the industry's private understanding of its product bore no resemblance to its public one. A memorandum written in 1963 by Addison Yeaman, the general counsel of Brown and Williamson, worked through the implications of the science and arrived at a sentence the company would spend the next three decades denying in public: "we are, then, in the business of selling nicotine, an addictive drug." And a strategy memo from within the same company in 1969 stated the entire method in a single line that has since become the most quoted sentence in the history of corporate deception: "Doubt is our product since it is the best means of competing with the 'body of fact' that exists in the mind of the general public." They were not confused about the science. They had decided, and written down, that confusion was the product they were selling.

The documents showed more than deception about cancer. They showed engineering. The companies understood that nicotine was the reason a smoker could not stop, and they had learned to manage its delivery. By treating the tobacco and the smoke with ammonia compounds, they could convert the nicotine into a freer form that crossed from the lungs into the blood and reached the brain faster and struck harder, a sharper dose and a firmer hook, a technique their

own documents discussed as "impact." A man who could describe this from the inside was Jeffrey Wigand, who had been vice president for research at Brown and Williamson until he was fired in 1993 after clashing with the company. Wigand agreed to tell what he knew on the CBS program 60 Minutes, and his account of it became a national ordeal. CBS, fearing a lawsuit from the industry for interfering with Wigand's confidentiality agreement, at first refused to broadcast the interview; the industry assembled a dossier attacking his character and circulated it; his marriage came apart under the pressure. The interview finally aired in February 1996, and the episode later became the subject of a film. What Wigand described, on television, was what the internal documents already showed on paper: that the men who swore the product was not addictive had spent their careers making it more so.

That oath was sworn in public, and it should be noted, because it is the moment the deception stepped into the open and dared the country to do something about it. On the fourteenth of April, 1994, before Williams's documents were public but as the pressure was building, the chief executives of the seven largest tobacco companies appeared together before a subcommittee of the House of Representatives chaired by Henry Waxman. One after another, under the lights and under oath, each raised his right hand, and one after another each testified that he did not believe nicotine was addictive. The photograph of the seven of them in a row, well dressed and composed, swearing to a thing their own laboratories had long since disproved, is the defining image of the industry, and not one of them was charged with perjury.

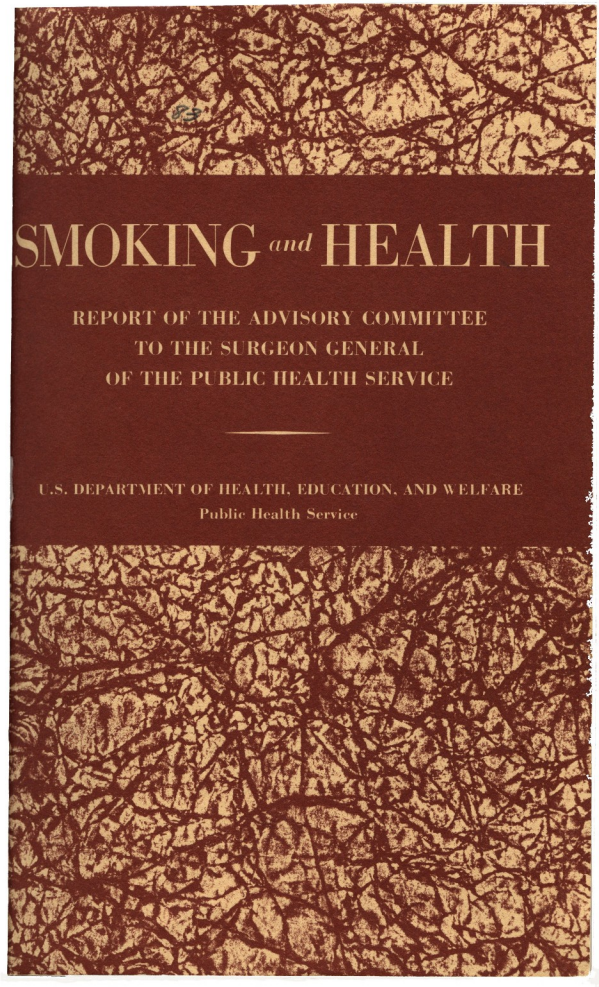
The lie did not stay a matter of shame. It became a matter of law, and the door had been opened two years earlier. In 1992, in *Cipollone v. Liggett Group*, the Supreme Court held that the federal warning label on the pack did not shield the companies from lawsuits alleging fraud, concealment, and conspiracy. That distinction mattered enormously: it meant a plaintiff could no longer be told that the warning label had

discharged the industry's duty, and that the deception itself, the fifty years of manufactured doubt, could be put in front of a jury. The states moved first. In 1994 the attorney general of Mississippi, Mike Moore, filed the first suit to make the companies repay what his state had spent through Medicaid treating dying smokers, on the theory that the cost of the industry's product should not fall on the taxpayer. Other states followed in a wave, and behind them came the federal government, which in 1999 brought a civil suit under the Racketeer Influenced and Corrupt Organizations Act, the statute Congress had written to break the Mafia.

The federal case was tried before Judge Gladys Kessler of the district court in Washington, and in 2006 she issued an opinion of more than sixteen hundred pages. She found, as a matter of fact and law, what she called a fifty-year scheme to defraud, and she catalogued its parts: that the companies had denied that smoking causes disease when they knew it did, denied that nicotine was addictive when they had engineered its delivery, denied that they marketed to children when their own plans show that they did, and denied that secondhand smoke was harmful against their own evidence. The Court of Appeals affirmed the core of it: a decades-long conspiracy to deceive the American public about the dangers of smoking, the addictiveness of nicotine, and the deliberate manipulation of its delivery. This is not a documentary's insinuation or an activist's charge. It is a finding of the United States courts, reached under the law written for organized crime, that the country's deadliest legal product had been sold for half a century by an enterprise the government proved had lied about it.

And here is the part that matters most: nothing was forbidden. The remedy Judge Kessler was permitted to impose was not prohibition, and in the end not even money, because the Court of Appeals had stripped away the multi-billion-dollar penalty she had sought. What remained was that the companies must publish corrective statements, admitting in their own advertising that they had lied and that their product kills. That was the sentence. A federal court

found a fifty-year fraud that had helped to kill millions of people, and the punishment was an apology in the newspaper, delayed by more than a decade of further appeal and finally printed in 2017. That apology, set against the sentence a young man receives for the sale of a plant that has killed no one, is the moral arithmetic in a single comparison.



Smoking and Health, the report of the Surgeon General's advisory committee, 1964.

Why, then, did it stay legal? The honest answer is the one we will keep returning to: because it enriched the corporate

elite, and because the parties who profited included the state. From 1938 until 2004 the United States ran a federal tobacco program, born in the Agricultural Adjustment Act, that guaranteed growers a minimum price and limited the supply to hold that price up. For sixty-six years the government was not a neutral referee of the tobacco business but a partner in it, with a financial stake in the crop's continued profit and a bloc of congressmen from the tobacco states to defend it. When the program was finally ended, in 2004, it was not abolished so much as bought out: the growers and quota holders were paid roughly ten billion dollars to let it go. The rules that did arrive over those decades came late and light, the warning label and the broadcast ban, and nothing resembling the authority to treat the cigarette as what it was. When the Food and Drug Administration at last tried, in the nineteen-nineties under its commissioner David Kessler, to assert jurisdiction over the cigarette as a nicotine-delivery device, the Supreme Court stopped it, ruling in 2000, in *FDA v. Brown and Williamson*, by a single vote, that Congress had never granted the agency that power. The nation's drug regulator was locked out of the nation's deadliest drug by an act of legislative silence, and did not gain the authority until 2009, forty-five years after the government's own Surgeon General had named the harm.

The harm, in the meantime, was steered. As the broad market shrank, the industry did with tobacco what a later chapter will show the food industry doing with sugar: it found the customers with the least power and marketed hardest to them. The historian Keith Wailoo, in his book *Pushing Cool*, documented how for sixty years the menthol brands were aimed, deliberately and in writing, at Black Americans, through the magazines they read, the neighborhoods they lived in, the jazz and community events their brands sponsored, and the free samples handed out on the corners of the inner city. Menthol cools and numbs the smoke, which makes the first cigarette easier for a beginner to take and the last one harder for anyone to give up, and today the great majority of Black smokers smoke menthol

while most white smokers do not. The disease and the death gathered where the marketing was aimed. The design was written into the law itself: when Congress finally gave the FDA authority over tobacco in 2009 and banned candy and fruit flavorings that appealed to children, it carved out one exception, the one flavor most used by Black adults, menthol, and left it on the shelf. When a regulator moved at last, in 2022, to close that carve-out and ban menthol, the proposal stalled and was delayed. The plant stayed legal for everyone; only the harm was sorted, by who had the least protection. That is the pattern of this whole thesis, running here inside one legal plant.

The largest reckoning tobacco ever faced was not a court but a settlement. Through 1997 and 1998, four states, Mississippi first among them, settled their Medicaid suits separately, and then in November 1998, in the Master Settlement Agreement, the companies agreed to pay the remaining forty-six states some two hundred and six billion dollars over twenty-five years. It was the largest civil settlement in American history, and it altered the arithmetic of the business without ending it. The companies raised their prices to cover the payments, which is to say that the smokers paid the settlement, and the states, now dependent on a stream of revenue that flowed only so long as people kept smoking, acquired a quiet interest in the survival of the industry they had sued. The most consequential thing the settlement produced was not the money. It was disclosure: the agreement forced millions of pages of internal documents into the open, where they were gathered into the archive now held at the University of California, San Francisco, on which this chapter, and much of the scholarship behind it, depends. The industry was compelled to surrender the evidence of its own conduct. It was not compelled to stop.

It did not stop, and it did not stay within tobacco. In the nineteen-eighties, as the writing on the wall grew harder to ignore, the two largest American tobacco companies bought the two largest American food companies. Philip Morris acquired General Foods in 1985 for some five and a half

billion dollars, and Kraft in 1988 for around thirteen billion, and folded them together into the largest food company in the country; through those firms it owned Maxwell House and Jell-O and Kool-Aid and Oscar Mayer and, later, the Oreo. R. J. Reynolds had already merged with Nabisco in 1985, and the buyout of the combined RJR Nabisco in 1988, the largest corporate takeover of its era, was chronicled by the reporters Bryan Burrough and John Helyar in *Barbarians at the Gate*. For the better part of two decades, the people who had engineered the cigarette and manufactured the doubt owned the American pantry, the cookies and the processed cheese and the packaged lunches and the powdered drinks, and the chemists and marketers who studied the science behind making a product irresistible moved from the tobacco divisions and into the food laboratories.

What they did with that knowledge belongs to a later chapter, and it must be stated with care, because the record supports a pattern and not a proven plot. Tera Fazzino, a researcher at the University of Kansas, and her colleagues, publishing in the journal *Addiction* in 2024, used those same disclosed tobacco documents to establish which food brands the tobacco companies had owned, and cross-referenced them against decades of nutrition data. They found that during the years of tobacco ownership, the food brands those companies owned were significantly more likely than others to be built to the point of maximum craving, the precise balance of fat and salt and sugar that the food scientist Howard Moskowitz, profiled in Michael Moss's *Salt Sugar Fat*, taught the industry to call the bliss point. The same infrastructure crossed over with the products: Kim Nguyen and Stanton Glantz, in the *American Journal of Public Health* in 2020, traced how Philip Morris transferred its methods for marketing to minority communities from its tobacco business directly into its food business. That the ownership overlapped is a matter of corporate record. That the products grew more compulsive under that ownership, and that the marketing playbook moved with them, are quantitative findings by named researchers working from

primary documents. Whether anyone sat in a room and resolved to addict the country to snack food as it had been addicted to cigarettes is not something the documents prove. What the documents do show is enough: the method that had kept the deadliest plant legal did not retire when the cigarette began to fall in popularity. It moved to the table.

At the end, the pinch of tobacco in the open hand returns. A plant a people had treated as sacred, offered sparingly and with words to the spirits, was taken up by an industry that made of it the deadliest product the country has ever sold, was found by a federal court to have lied about it for fifty years. And, they still to this day, sell it. At no point in that history was the plant forbidden, because at every point it generated enormous profit, and the parties it paid included the government that might have stopped it. That is the measure this book will hold against every plant that was forbidden. The coca leaf an Andean farmer has chewed for four thousand years, the small cactus a church takes as its sacrament, the hemp that once rigged the ships and the poppy that once eased the dying: not one of them carries a death toll within sight of the plant we have never thought to ban. The question is not why they were forbidden. It is why this one was not. And tobacco was not the only plant the country chose to keep. The next chapter turns to corn, a sacred grain of that same world, distilled into a second legal killer that the country has spared for the same reason it spared tobacco.

Chapter Two

Alcohol



The last chapter began with a pinch of tobacco. This one begins with a handful of corn.

In *The Forgotten Harvest* the corn was *mandaamin*, the sacred gift, the plant that came to the people in the old accounts as a living relative and fed them through every winter of the record. It was the center of the three sisters, the staple that made the villages of the Great Lakes possible, the plant, the grain a family dried and banked and planted again. No plant was held in higher regard.

It was also far older on this continent than the people who would later exploit it. Maize was domesticated in the highlands of Mesoamerica some nine thousand years ago, bred up from a wild grass over a span of human effort longer than most written history, and it reached the Great Lakes centuries before any European did. By the time the first traders arrived, corn was not a novelty but the settled foundation of a working agricultural science, and it did more than feed a household. It was the crop that bound communities together and moved along the trade routes, the good that a nation with a surplus could carry to one without, the thing that made alliance and exchange possible. The corn kept the people fed, and the trade in it helped keep them one people.

What undid that order was, in part, a collision of laws that one side did not know it was having. Indigenous land in the Great Lakes was held in common; it belonged to the nation and the coming generations, and no single man, whatever his standing, held the authority to sell it out from under them. The European idea that a mark on a piece of paper could sever a people from their ground forever was not merely unwelcome; it was, to many who signed, unintelligible. The record holds instances of cessions obtained from men who had no power to grant them, of agreements whose permanence the signatories did not grasp, and of liquor poured to ease the signing. This is a documented pattern, trader by trader and council by council, and it should be called a pattern and not a plot. It did not require a single

guiding hand. It required only a steady asymmetry of knowledge, and a bottle.

The contrast is stark: a plant that kept the Indigenous alive over many centuries was then, in several generations following colonialism, transformed into a product to corrupt a people's spirit. Corn is the chief feedstock of modern American alcohol and industrial ethanol. The corn that once fed the people is now distilled, at industrial scale, into the thing that helped break them. This is the whole pattern in a single crop: a sacred Indigenous plant turned against its own origin. And it sets up a paradox even sharper than tobacco's, because alcohol is the one intoxicant in these pages that America actually did ban, and then chose to un-ban.

The same arithmetic is the test that sorts every plant in these pages. The Centers for Disease Control and Prevention attributes roughly one hundred and seventy-eight thousand American deaths a year to excessive alcohol use. In 2023 the World Health Organization stated plainly that no level of alcohol consumption is safe, and the International Agency for Research on Cancer classifies alcohol as a Group 1 carcinogen, the same category as tobacco, asbestos, and radiation. By the measure of harm, alcohol belongs with the deadliest substances a person can buy. And it is sold on every shelf in the country, taxed but never forbidden.



"The Drunkard's Progress," Nathaniel Currier, 1846.

Except that once it was. Alcohol is the control experiment for this entire study: the same category of harm as the plants that follow, and the opposite legal fate. Between 1920 and 1933 the United States banned it outright, wrote the ban into the Constitution itself, found the ban unworkable, and repealed it inside thirteen years. No other constitutional amendment has ever been undone. That reversal stands beside the bans on the plants of the coming chapters, which fell on communities with less power and ran, in some cases, for a century and are running still. A white-majority drug's prohibition was reversed in thirteen years. The question is why, and the answer is not that alcohol turned out to be safe.

The record shows what was done to the nations of the Great Lakes. Alcohol was not only a product sold to Indigenous people; it was an instrument used against them, and the documentation is in the traders' own ledgers and the government's own treaties. The historian Peter Mancall, in *Deadly Medicine: Indians and Alcohol in Early America*, traced how the fur trade ran on liquor: advanced on credit,

poured to open a bargain, and used to strip furs that should have paid an earlier debt. It was a lever for breaking down a self-sufficient economy that had little use for imported goods, a method close enough to the opium the same era's empires were pressing on China that the comparison is Mancall's, not an embellishment. This is a documented pattern, and it should be called a pattern and not a plot: the record proves the practice, trader by trader and treaty by treaty, without needing a single coordinating hand behind it.

The government's part is on the record too, and it is self-incriminating. While Washington moved to ban everyone else's whiskey in Indian Country, it reserved liquor for its own dealings, and the proof sits in the treaties. When the Choctaw ceded their land at Doak's Stand in 1820, the treaty empowered the resident federal agent "to seize and confiscate all the whiskey which may be introduced into said nation," with one exception: the whiskey "used at public stands, or brought in by the permit of the agent, or the principal Chiefs." The same government that criminalized the trader's bottle licensed its own. The Ojibwe writer George Copway, among the first Native authors to publish in English and a temperance voice of his generation, named what came across that frontier without flinching. What the whites brought, he wrote, were "traders, spirit-sellers, horse thieves, counterfeiters," men "willing to raise themselves by the downfall of the Indian race," and of the whole account he drew the shortest possible verdict: "What war has not done, strong drink has." The nations named the harm themselves, at the treaty table, as the Choctaw did in council in 1801 when they asked that the strong drink the commissioners had carried be kept away while they signed away their ground.

Then the government named the harm in law, and the law failed. A restriction in 1802 gave the President authority to restrain the liquor trade among the tribes. An act of 1832 declared that no ardent spirits should be introduced, under any pretense, into the Indian country. An act of 1834 made supplying liquor a crime. The Supreme Court spent the next

half-century working out how far this power reached, first holding in *Matter of Heff* in 1905 that a Native person who had become a citizen was beyond the ban, then reversing itself in *United States v. Nice* in 1916 to rule that even Native citizenship did not exempt a person from federal liquor prohibition. Enforcing the ban became its own machinery: by 1900, liquor violations in Indian Country accounted for something like one in twenty of all federal prisoners. That prohibition was not lifted until 1953, when Congress finally let each tribe set its own liquor policy. The contradiction is plain: federal alcohol prohibition in Indian Country lasted one hundred and twenty-one years, nearly ten times the length of national Prohibition, and it did not stop the harm, because the drivers of the harm were never only the bottle.

Which brings us to the most delicate point in the chapter, and the one that must be handled exactly. The disparity is real and it is large: the CDC's data put the alcohol death rate among American Indian and Alaska Native people at roughly four times the rate for white Americans. For a century that gap was explained by a story about Native bodies, the so-called firewater myth, the claim that Indigenous people are biologically more vulnerable to alcohol. That claim is false, and it has been tested. The researcher Garcia-Andrade and colleagues, publishing in the *American Journal of Psychiatry* in 1997, measured the response to alcohol in people of Native heritage and found that those with more Indigenous ancestry reported, if anything, less intense effects at the same blood-alcohol level, the opposite of what the stereotype predicts. There is no genetic firewater. The physiology does not support the story that was told about it for a hundred years.

What the myth does support is an evasion. If the cause of the disparity lives in Native blood, then it lives nowhere that policy has to answer for, and the real drivers go unnamed. Those drivers are the ones every honest account records: dispossession, the imposed poverty that followed it, the trauma carried across generations, and the sheer distance to a clinic in the places Native people were left. The belief is not

even inert. In the clinical literature it has been linked to worse treatment for Native patients, which means the myth does a second injury on top of the first. It is the racialized cover story that lets the structural causes off the hook, and the goal here is to separate intention from the unconscionable objective facts.

If the harm was steered downhill, the product was laundered upward, by the same move tobacco used. For a long time alcohol was sold as medicine, prescribed for everything from snakebite to heart disease. During national Prohibition that medical channel stayed open while every other one closed: the Treasury issued numbered prescription pads, physicians could legally prescribe up to a pint of whiskey per patient every ten days for ailments as vague as influenza and depression, and Americans filled something on the order of eleven million such prescriptions a year. It was lucrative enough to build an empire on. The drugstore chain Walgreens grew from roughly twenty stores in 1920 to some five hundred and twenty-five by the decade's end, a boom the whiskey trade did much to feed. After repeal came the advertising. The United States Brewers Foundation ran a long campaign through the nineteen-forties and fifties called "Beer Belongs," more than a hundred magazine advertisements in titles like McCall's and Collier's, each a commissioned painting of wholesome American family life with a glass of beer set inside it, repositioning an intoxicant as respectability itself. And for decades afterward the public was told that moderate drinking was good for the heart, a belief the WHO's 2023 statement retired. Part of that reassuring science turns out to have been bought. In 2018 the National Institutes of Health shut down a hundred-million-dollar trial on moderate drinking after an internal investigation found that its own officials had solicited the funding from the alcohol industry, five global producers among them, Anheuser-Busch InBev, Carlsberg, Diageo, Heineken, and Pernod Ricard, who were to supply the bulk of the cost, and that the study had been shaped to be likely to find the answer the sellers wanted. When the science that

sells a product is paid for by the seller, moderation-is-healthy is marketing in a lab coat.

AN ACT TO REGULATE TRADE AND INTERCOURSE WITH THE INDIAN TRIBES

> If any person shall sell, exchange, give, barter, or dispose of any spirituous liquor or wine to an Indian, in the Indian country, such person shall forfeit and pay a sum not exceeding five hundred dollars.

Act of June 30, 1834.

Now the inconvenient truth, the fact that makes alcohol the control experiment. In January 1920 the Eighteenth Amendment took effect and the manufacture, sale, and transport of intoxicating liquor became unconstitutional. It did not, however, ban private possession or consumption, and that gap was a door for anyone with money and foresight: a household that had stocked its cellar before the deadline could keep drinking from it, legally, for as long as the supply held. The ban did not fall evenly in any other respect either. The historian Lisa McGirr, in *The War on Alcohol*, showed that enforcement came down hardest on immigrants, Black Americans, Mexicans, and the poor, while the wealthy drank from those legal cellars and behind the doors of private clubs. It was, in the phrase of the period, a law for the poor and a joke for the rich. And when it proved unenforceable and ruinous, the country reversed itself with unusual speed. In the spring of 1933 the Cullen-Harrison Act legalized beer of a modest strength, a first crack in the wall; that December the Twenty-first Amendment, sent to the states by the Blaine resolution and ratified not by the legislatures but by special state conventions, repealed the Eighteenth outright, the only such reversal in the history of the Constitution. McGirr's deeper point is the one the chapters ahead will carry forward. Prohibition did not simply end; it built something. The alliance of moral reform and federal power, she writes, produced "a dramatic expansion of federal policing and an increase of state and local policing in the quasi-military sphere of crime control." That machinery,

and the public appetite for it, is what the war on the coming plants would inherit.

Set side by side, the two histories diverge. A white-majority drug was banned, the ban was enforced against the people with the least power, it was found intolerable to the majority who drank, and it was undone inside thirteen years. The plants of the next chapters, associated in the public mind with Mexican, Black, Chinese, and Indigenous communities, were banned and stayed banned, some of them for the better part of a century. The difference was not danger. Alcohol is a Group 1 carcinogen, a known inhibitor of self-control: a destroyer of family bonds and domestic harmony. The difference was who drank it and who profited from it.

Who pays the cost of a policy and whose stock rises are decided, again and again, by who holds the economic power. From the fur companies that governed the early trade to the corporations that lobby the modern Congress, commercial interest has sat closer to the writing of the rules than the people those rules were meant to protect. The agencies built to stand between the public and private greed have, in case after documented case, been turned to the service of profit: the tobacco program that made the government a partner in the crop, the alcohol trial the industry was caught funding, the science and the medical authority bent to sell the product. Not every time, but often enough to be the pattern and not the exception, policy has favored the seller while the health of the population declined.

The CDC's hundred and seventy-eight thousand deaths a year come with an economic cost the agency has put in the hundreds of billions of dollars, more than a third of it borne by government, which is to say by the public, on behalf of a private and profitable harm. American Indian and Alaska Native communities carry roughly four times the white death rate, for the structural reasons named above and not the mythical one. And around the legal harm a second industry has grown. The American Medical Association reclassified alcoholism first as an illness, in 1956, and then as a disease,

in 1966, and that medical reframing, whatever its clinical merits, opened a market: an alcohol-treatment economy now worth tens of billions of dollars a year, selling the cure on the back end of a product that stays legal throughout. Two profit centers, one legal harm.

There is a line from the study of systems, coined by the cybernetician Stafford Beer: the purpose of a system is what it does. Judge it by what it produces, not by what it says it intends. In our system the seller collects, and so does the treater, and in this way the profiteers subverted the life-giving corn plant and degenerated it into a poison. This is the structural pattern of our system, not a proven conspiracy of enfeeblement, and the pattern is clear enough.

Alcohol, then, was spared not because it was safe. It was spared because it was the majority's drink and because it paid, the government and the elite ruling class among those it paid, exactly as tobacco was. It is the plant that proves the rule by being the one exception the country could not stomach living without. The plants in the chapters that follow had no such protection. They were the ones grown by the wrong people, or smoked in the wrong neighborhoods, or carried across the wrong border, and forbidden for it. The first is cannabis, a plant far less deadly than the one just described, and outlawed all the same.

PART TWO

The Forbidden Plants



Chapter Three

Cannabis

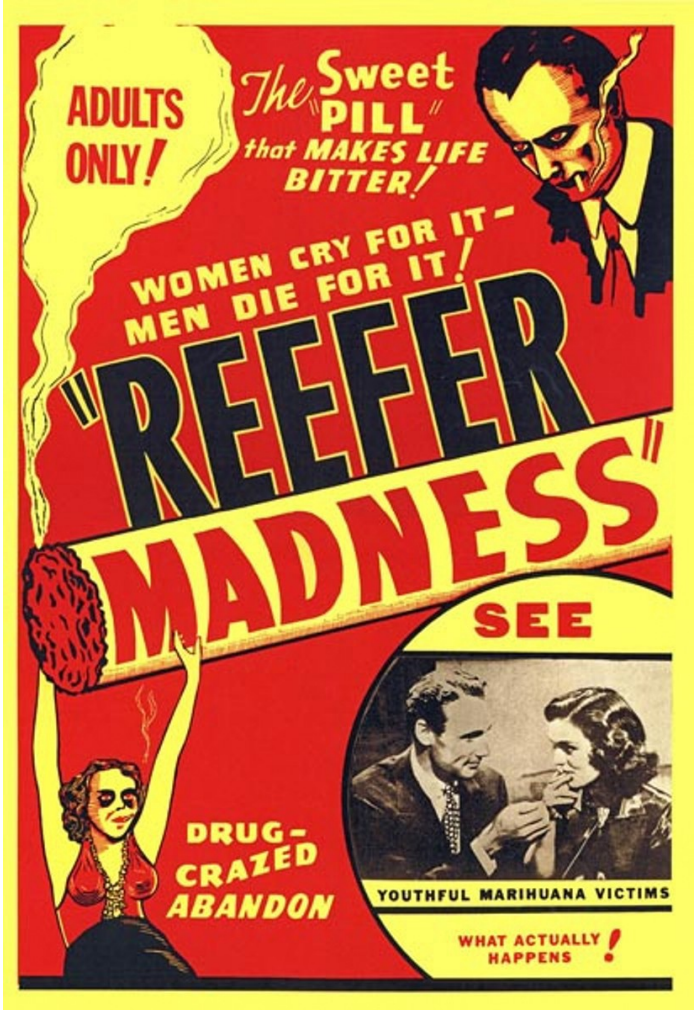


The first two chapters were about plants that kill and are sold anyway. This one begins with the plants that do far less harm and were forbidden regardless. If tobacco and alcohol are the control experiment, cannabis is the first result the experiment predicts: a plant near the bottom of any honest ranking of danger, outlawed at the top of the country's list, and used, for the better part of a century, to put hundreds of thousands of people in cages.

In America the plant did not start as a drug at all. Hemp, the same species grown for fiber, was one of the indispensable crops of the early modern world. It made the rope, the rigging, and the sailcloth that moved every navy and merchant fleet on the ocean, and no seafaring power could do without it. When the English planted Virginia, they did not merely permit hemp; in 1619 the Jamestown colony required settlers to grow it. For three centuries the plant was strategic infrastructure, cultivated in Virginia and Kentucky and prized in London. Even setting aside the folklore that has grown up around this, the claim that the Declaration was written on hemp, that paper currency was made from it, or that some exact percentage of all sails were hemp, the plain documented fact is still remarkable: the plant later branded an intolerable menace was, for most of American history, a crop the government ordered farmers to raise. A plant that foundational does not become a felony because of anything in its chemistry. The explanation has to be looked for somewhere other than the plant.

It is found, first, in a word. The plant Americans had long known as cannabis, or Indian hemp, and encountered mostly in the tinctures on a pharmacist's shelf, acquired a new and foreign-sounding name in the early twentieth century: marihuana. To later generations, marijuana is the most common modern spelling. The renaming to marihuana was not innocent. It arrived with Mexican migration across the southern border, and it let a familiar medicine be recast as an alien intoxicant carried by an alien people. The earliest bans followed that fear rather than any science. Mexico itself prohibited cannabis in 1920, and a string of American states,

Massachusetts in 1911, Maine in 1913, Vermont in 1915, and much of the West and Southwest, banned it well before any federal law and well before any national panic. The historian Isaac Campos, in *Home Grown*, traced how the idea of the drug as a cause of madness and violence crossed the border in both directions. The point to hold, because it governs the chapters that follow, is a careful one. Cannabis is not a New World or an Indigenous plant; its origins are in Asia. What was racialized was not the plant but the ban, aimed from the beginning at the people associated with it.



Poster for the film Reefer Madness, 1936.

The ban went federal when a bureau needed a reason to exist. National alcohol Prohibition ended in 1933, and the Federal Bureau of Narcotics, created in 1930 under Harry Anslinger, faced the loss of the enemy that justified its budget. Anslinger found a replacement in marihuana, and the campaign he ran for it was explicitly racial: his public statements and his files tied the drug to Mexican immigrants, to Black Americans, to Filipinos, and to jazz musicians, and to a violence the evidence never supported. The popular culture supplied the mood, the 1936 film *Reefer Madness* chief among the propaganda. He was not a lone author of any of this; the state bans preceded him and a broad network of reformers stood behind him. But he was the federal engine, and the incentive that drove him was bureaucratic before it was moral. A version of the story suggests the real hand behind prohibition was industrial, a plot by DuPont and the newspaperman Hearst to kill hemp for the sake of nylon and wood-pulp paper. That story is popular and it is wrong. The article usually cited as the trigger ran six months after the law passed; the machine supposedly threatened had never reached the market; the states that banned cannabis first had no hemp industry to suppress. The instinct that money and power drive drug law is sound, and everything here proceeds from it, but in this case the evidence points not to a paper trust but to a bureau protecting itself and the controlled psychological manipulation of a country into fear of a minority. Fear-driven narratives like this one recur throughout history, and in the twenty-first century they have become so frequent as to seem almost banal. The playbook of those in charge, from time immemorial, appears to follow an obvious pattern. But in truth we can only guess as to why their motivations seemed designed to manipulate populations through fear in order to achieve their ends.

When the law came, the doctors did not immediately bow to the narrative. The Marihuana Tax Act of 1937 did not ban the plant outright; it wrapped it in a prohibitive scheme of taxes and transfer stamps whose real purpose was to criminalize the trade. At the hearings, the American Medical Association

sent its legislative counsel, Dr. William Woodward, to object, and he objected on the ground that matters most here: that there was no evidence marihuana was a dangerous drug, and that no such evidence had been gathered from the agencies that would have studied it. He was treated as an obstacle and brushed aside, and the Act passed and took effect on the first of October, 1937. The leading medical body in the country said the danger had not been shown, and the law was written as though it had. Legality and danger often come apart on the public record; in this case the whole argument was declared fact in a committee room, and the results are what the system desired.



Harry J. Anslinger, head of the Federal Bureau of Narcotics.

The scheme held for three decades, and then what replaced it was even worse. In 1969, in *Leary v. United States*, the Supreme Court struck the Tax Act down. The law had required anyone who transferred marihuana to register and buy a federal tax stamp, but a person could not obtain that stamp without formally declaring to the government that he dealt in the drug, which in every state that had already outlawed it meant confessing to a crime. To obey the tax was to incriminate yourself, and the Court held that this violated the Fifth Amendment. Congress answered the next year with

the Controlled Substances Act, which sorted every drug into schedules and placed cannabis in Schedule I, the category reserved for substances with high abuse potential and no accepted medical use, alongside heroin. It was meant to be temporary, a placeholder pending study. The study came. In 1972 the commission President Nixon had himself appointed, chaired by Raymond Shafer, reported in *Marihuana: A Signal of Misunderstanding* that personal possession should be decriminalized. Nixon rejected the findings of his own commission, and cannabis stayed in Schedule I, where it remains. This is the original sin of the drug schedules: not an oversight, but a documented decision to override the government's own evidence, and to keep overriding it for fifty years.

Who paid for that decision is the part the numbers make undeniable. Cannabis is used at roughly the same rate by Black and white Americans, and its harms, disputed but clearly milder than those of alcohol, opioids, or tobacco, fall on users without regard to race. The collapse of family structures that follows arrest and incarceration does not. In *The War on Marijuana in Black and White*, the American Civil Liberties Union counted some eight million marijuana arrests between 2001 and 2010, nearly nine in ten of them for simple possession, and found that a Black American was roughly three and a half times more likely than a white American to be arrested for it, on identical rates of use. States spent on the order of three and a half billion dollars a year enforcing that gap. A follow-up in 2020 found the disparity essentially unchanged, present in almost every county, and persisting even in states that had begun to legalize. The plant did not become more dangerous or less dangerous across those decades. Only the enforcement moved, and it moved downhill, onto the same communities every chapter that follows will find at the bottom of the sorting.

And now, slowly, the plant is being legalized. The 2018 Farm Bill legally severed hemp from marihuana again, defining the fiber crop back into legitimacy, and in 2024 the Drug

Enforcement Administration proposed at last to move cannabis out of Schedule I. The reversal, when it is complete, will prove the same thing the ban did: that the plant's legal fate was never a reading of its danger. Cannabis is not an Indigenous plant, but the structures built to forbid it, the bureau, the schedule, the possession arrest, are the same machine that transformed Indigenous nations and then perverted their sacraments, and they are the beginning of the playbook later used to profit the industries that harm people. And as the next chapter will show, the technique of turning fear into profit, and of making harm out of what was once good, is an enduring one. This time the plant is Indigenous, chewed on this hemisphere for thousands of years, and it too would be taken up by the same governments and corporations and transformed.

Chapter Four

Coca



Of all the plants discussed in these pages, none demonstrates the hypocrisy of our regulatory systems better. When two groups of people utilize the same leaf on the same morning, but one group is deemed a criminal and the other is called an industry, we have a clear divarication of reason.

In the Bolivian Chapare a farmer folds a wad of coca leaves into his cheek with a pinch of lime and mineral ash, the way the people of those mountains have done for as long as there have been people in them. It is not intoxicating; the effect is harder to discern than a cup of coffee. It quiets his hunger, steadies him against the thin air, and readies him for a day of work, so that everything he does seems just a bit clearer and a bit easier. And yet, by the letter of an international treaty his own government signed, the quid in his cheek is contraband, a practice the nations of the world formally agreed to abolish.

That same week, a sealed shipment of the identical leaf clears United States customs and travels to a chemical plant in New Jersey, legally, under a federal license held by a single American company. There the leaf is stripped of its trace alkaloid, the flavoring is shipped south to Atlanta, and the cocaine drawn out of it is sold into the medical market. Nothing about the transaction is hidden, and nothing about it is against the law.

The same plant is a felony in an Andean cheek and a licensed trade good in an American factory. The distance between those two outcomes shows, at the very least, that a disparity has been created within our legal and regulatory systems. We can be quite certain the leaf was not banned for its danger, because to take coca and convert its alkaloid into cocaine is comparable to purifying the caffeine out of coffee or the morphine out of a poppy: each would be an inappropriate, even malicious, treatment of the plant rather than an ordinary use of it. Yet for years cocaine itself was an ingredient in American beverages and a recognized addiction within the early medical profession, prescribed with abandon and condoned much as the cigarette had been. Was the harm

ever weighed against who was using it? Chewing the leaf alters the mind less than a cup of coffee does. Cocaine, by contrast, has clearly destroyed lives. Perhaps the responsible agencies meant to protect Indigenous populations, as they claimed to be doing when they banned alcohol in Indian Country. But there, as the last chapter showed, they did the opposite of protecting anyone. The answer is once again a profit motive.



A page from the Album Mariani, the coca wine's gallery of celebrity endorsers, c. 1901. Among them the aviator Santos-Dumont, who called Vin Mariani "the best accumulator of human energy." Each notable thanks "the kind inventor of Coca."

That distinction has to be stated plainly, because everything here depends on it. The coca leaf is not cocaine. The leaf carries the cocaine alkaloid at roughly one percent of its dry weight, released slowly when it is chewed with an alkaline pinch, and its effect is a mild, steady lift comparable to coffee, along with relief from hunger, fatigue, and altitude. Cocaine is that alkaloid stripped out and concentrated, a fast and genuinely addictive drug. The argument of this chapter is not that cocaine is harmless. It is that the leaf is not the drug, and that of the two, only the leaf-chewing Indigenous population was criminalized.

The leaf itself is old. Coca, the plant botanists call *Erythroxylum coca*, has been used in the Andes for as much as four thousand years, its quids and lime-gourds turning up in the archaeological record long before the Inca. Under the Inca it was distributed to laborers, carried by the running messengers who held the empire together, and offered to the earth and the mountain spirits. Then the Spanish came, and after an initial burst of clerical hostility the crown discovered coca's usefulness and taxed it, because the leaf kept Indigenous laborers working the silver mines of Potosí that made Spain rich. From the very beginning, as the historian Paul Gootenberg documents in *Andean Cocaine*, the leaf's legal standing tracked its usefulness to whoever held power over it. The purpose of a system, in the cybernetician Stafford Beer's phrase, is what it does.

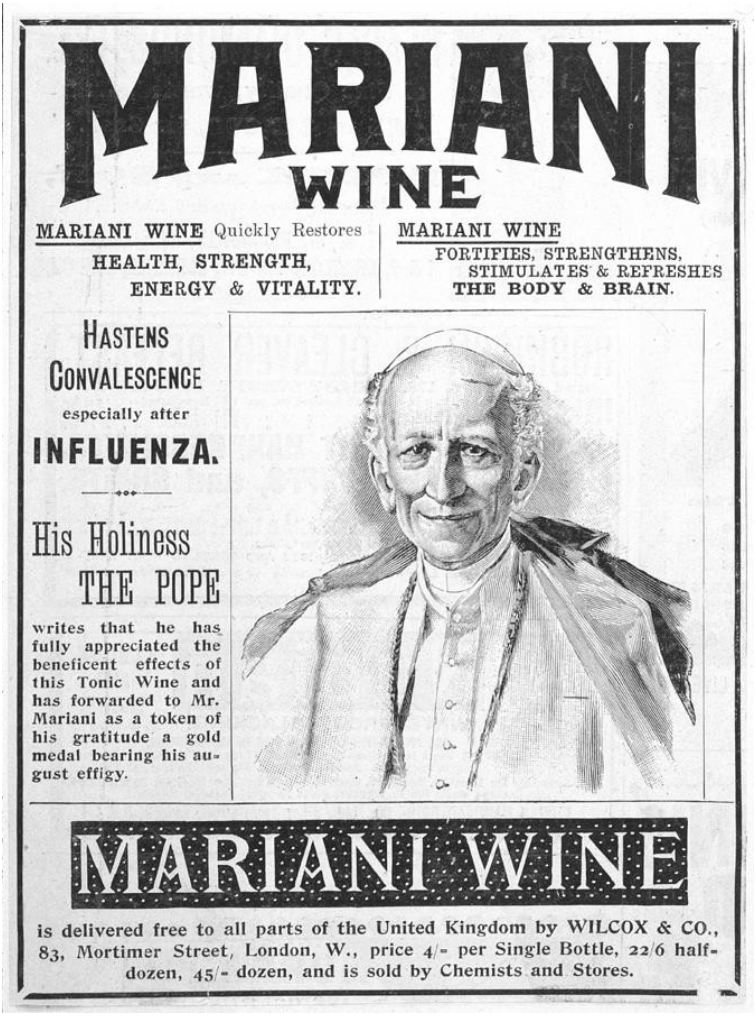
In 1860 the drug was separated from the plant for the first time, when the chemist Albert Niemann isolated the cocaine alkaloid in a German laboratory. A wave of commercial products followed, all built on the concentrated extract: coca wines, tonics, and patent medicines, and in 1886 an Atlanta druggist named John Pemberton mixed coca extract and kola nut into the syrup he called Coca-Cola. That commercial enthusiasm was then turned into a public panic, and the panic was deliberately racial. Around the turn of the century the American press filled with false warnings tying cocaine to Black Americans, a scare with no basis in fact and enormous consequences in law. Coca-Cola, as the historian Bartow

Elmore recounts in *Citizen Coke*, quietly switched around 1903 to a decocainized coca extract, keeping the flavor and removing the drug. And when the United States passed its first great drug-control law, the Harrison Narcotics Act of 1914, that law taxed and controlled coca and opium but wrote in an exemption for "decocainized coca leaves or preparations therefrom." That corporate exemption was enacted into American law before the global prohibition of the leaf had even begun.

The prohibition, when it came, was justified by a commission that had made up its mind in advance. In 1949 the United Nations sent a Commission of Enquiry on the Coca Leaf to the Andes, chaired by Howard B. Fonda, a senior executive of the pharmaceutical firm Burroughs Wellcome. Before the study had properly begun, Fonda told a Peruvian newspaper what he expected to find: that coca chewing was "not only definitely harmful and deleterious, but is the cause of the racial degeneration of many population groups." Meanwhile the stimulants the powerful preferred, coffee and alcohol and the cigarette, were promoted openly in every form of advertising and art and set upon the western kitchen table. The commission's 1950 report duly recommended that coca chewing be phased out, even as it conceded, in its own text, that the practice "does not constitute an addiction, but a habit." The verdict came first and the evidence was fitted to it, colonial contempt in the language of public health.

That report became the foundation of a treaty. The 1961 Single Convention on Narcotic Drugs placed the coca leaf in Schedule I, the most restricted category, alongside cocaine and heroin, and two of its articles, set side by side, show the contradiction plainly. Article 49 required that coca-leaf chewing "be abolished within twenty-five years" of the Convention taking effect, which set a deadline, 1989, for the legal extinction of a practice four thousand years old. Article 27 permitted the coca leaf to be used "for the preparation of a flavouring agent, which shall not contain any alkaloids," which is precisely and only what Coca-Cola does with it. In two numbered articles a few pages apart, the same treaty

ordered the Indigenous chewer's custom abolished and licensed the American corporation's use of the identical plant.



Pope Leo XIII, who awarded Mariani a Vatican gold medal, on a Vin Mariani advertisement. A reigning pope lent his face to a cocaine tonic; within a generation the same compound sat in the law's highest tier of prohibited drugs.

The corporation's end of that arrangement is not a secret, which makes it stronger than any secret. The decocainizing is done at a plant in Maywood, New Jersey, run for decades by the Maywood Chemical Works and later by the Stepan

Company. In 1988 the New York Times reporter Clifford May laid the chain out in an article titled "How Coca-Cola Obtains Its Coca": Stepan legally imports coca leaf, produces the decocainized flavor extract that Coca-Cola knows internally as Merchandise No. 5, and sells the cocaine it removes to the pharmaceutical manufacturer Mallinckrodt for medical use. One company, holding the sole federal license to import the leaf, processes a reported hundred tons or more of it a year. The exact figure is held in Drug Enforcement Administration quota notices and is best given as a reported range, but the fact needs no precise number. The exemption the treaty denies to the Andean farmer, it grants to a soft-drink company.

The farmer, meanwhile, got the other end of the law. For roughly three decades the United States financed forced eradication of coca in Bolivia's Chapare and the growing regions of Peru and Colombia, and under Plan Colombia it paid for aerial spraying of the herbicide glyphosate over the countryside, roughly one and three-quarter million hectares of it through 2015. The spraying killed food crops along with coca and drew years of health, environmental, and displacement complaints from campesino, Indigenous, and Afro-Colombian communities. Colombia suspended the aerial program in 2015 after the World Health Organization's cancer agency classified glyphosate a probable human carcinogen. Through all of it, the burden of enforcement fell on the growers, and never on the licensed importer in New Jersey.

The people who paid, in the end, forced the question onto the table. A coca-grower union leader named Evo Morales rose to national politics on the slogan "Coca sí, cocaína no," coca yes, cocaine no, and in 2006 became the first Indigenous president in Bolivia's history. Bolivia withdrew from the 1961 treaty and rejoined it in 2013 with a formal reservation protecting the chewing and cultivation of coca in its natural state on Bolivian soil, and the reservation held over the objection of the United States, which to this day makes it illegal for an ordinary consumer to import the leaf. Then, in

2025, the World Health Organization's expert committee completed the first genuine scientific review the leaf had ever received, and found no evidence of meaningful harm to public health and no significant potential for dependence. Having found that, the committee recommended keeping coca in Schedule I regardless, on the reasoning that the leaf can be processed into cocaine. The science exonerated the plant, and the classification survived on account of the drug the plant can be made into.

So it returns to the farmer and the factory. The evidence now says what the Andes always knew, that the leaf is mild and that its traditional use does no measurable harm. The law says what it has said since a pharmaceutical executive toured Peru with his conclusion already written: that the same leaf is a controlled narcotic in the mouth of the farmer who grew it and a lawful flavoring in the hands of the company that buys it. Nothing about the plant changed across that century. Only the hands it passed through, and the power behind them, decided which use was a crime. The next chapter turns to another Indigenous sacrament, a small desert cactus whose people had to defend it not at a treaty table, as the growers of coca did, but before the Supreme Court of the nation that had taken their land.

Chapter Five

Peyote



The ceremony lasts all night. Inside a tipi raised for the purpose, around a crescent of packed earth and a fire tended until dawn, members of the Native American Church pray, sing, and take the sacrament: buttons of a small desert cactus, eaten or brewed into a tea and passed around the circle with care. A water drum and a gourd rattle carry the songs from one person to the next. At first light the meeting closes with a simple breakfast of water, corn, fruit, and meat. This is worship, nearer to a communion service than to anything the word "drug" brings to mind, and it belongs to the oldest religion native to this continent still in regular practice.

The sacrament is peyote, a spineless cactus that botanists call *Lophophora williamsii* and that grows only in the Chihuahuan Desert of northern Mexico and a strip of southern Texas. It contains the hallucinogen mescaline. Once again, two documented facts about peyote point to the same playbook used to deny a people the use of a sacred plant. It is a plant that causes no physical harm, does not destroy families, and in fact binds a community and strengthens its spiritual bonds. Yet our leaders decided they knew what was good for that community's health, and by acts of legislation defined what was bad. The facts speak for themselves. First, the ceremonial use is ancient; Congress itself found in 1994 that the sacrament "has for centuries been integral to a way of life." Second, peyote is not addictive. Mescaline produces no physical dependence and no withdrawal, and the body builds tolerance to it within a few days, which discourages repeated use rather than driving it. Peyote is a potent hallucinogen with real effects, and no honest account calls it harmless to the psyche. The exploration of consciousness carries its own dangers, and traditional practice hedges it with warnings and protocols, but by every measure of dependence and public harm peyote sits far below alcohol or tobacco. Yet the United States files peyote in the same legal category as heroin.

The suppression of peyote did not begin with a legislature or a court. It began with a federal agent. From 1888 the United

States agents at the Fort Sill reservation ordered peyote forbidden in any form, and the Bureau of Indian Affairs pressed against it for decades afterward. Missionaries and assimilation reformers joined the effort, because the ceremony was Indigenous religion and the erasure of Indigenous religion was the aim. A wave of state bans followed, beginning in Oklahoma Territory in 1899. The Comanche leader Quanah Parker and the Caddo and Delaware leader John Wilson, the two founding figures of the modern peyote religion, defended it, and Parker testified against an Oklahoma ban before his death in 1911. When a bill to outlaw peyote nationally reached Congress in 1918, the peyotists answered by incorporating, that October, the Native American Church, which gave their religion a legal name and a measure of protection. They had to make their faith a chartered corporation to keep the government from banning it.

For the next seventy years the church fought a defensive war. Its members lobbied down one proposed federal ban after another. When Congress passed the Controlled Substances Act in 1970, it placed mescaline and peyote in Schedule I, the tier reserved for drugs deemed to have high abuse potential and no accepted medical use, and a Drug Enforcement Administration regulation carved out a narrow exemption for bona fide ceremonial use. That exemption was an administrative allowance, not a right, and what an agency grants an agency can take back. The strongest protection of the era came instead from a state court. In *People v. Woody*, in 1964, the California Supreme Court overturned the peyote convictions of a group of Navajo church members. Writing for the court, Justice Tobriner described plainly what the plant was to them: peyote, he wrote, "is the sole means by which defendants are able to experience their religion," so that "to forbid the use of peyote is to remove the theological heart of Peyotism." Weighing the state's interest against that religious need and finding the state's the weaker, he added a sentence that reaches well past peyote: "In a mass society, which presses at every point toward conformity, the protection of a self-expression, however unique, of the

individual and the group becomes ever more important." For a generation, that balancing test was the shield.



Peyote (Lophophora williamsii).

Then a peyote case took the shield away, and not only from the peyotists. Al Smith and Galen Black, both members of the Native American Church, were fired from their jobs as drug and alcohol counselors in Oregon for taking peyote in a ceremony, and were then denied unemployment benefits. Their case reached the Supreme Court, and in 1990, in *Employment Division v. Smith*, the Court ruled against them. Writing for the majority, Justice Antonin Scalia held that the First Amendment does not exempt a person from a neutral law that applies to everyone, even when that law forbids a sacrament, because to allow such exemptions would make each believer, in his phrase, "a law unto himself." That majority opinion became the binding law of the land. Three of the nine justices disagreed with it. One of them, Justice Harry Blackmun, wrote a dissent, which is the opinion of a judge on the losing side; it carries no force of law

but records why he believed the Court was wrong. Peyote, Blackmun wrote, "has nothing to do with the national drug problem." The ruling discarded the compelling-interest test that had protected religious minorities for decades. And here is why a case about a single Native American sacrament reached every faith in the country: the Court chose not to decide it narrowly, on peyote alone, but to announce a broad new rule, that a neutral law overrides any religious objection. A case brought by a few thousand peyote worshippers had reset the limits of religious freedom for everyone.

The reaction was wide, because the decision had weakened every faith's protection, not only that of the peyotists. Three years later, in 1993, Congress passed the Religious Freedom Restoration Act to restore by statute the strict test the Court had thrown out. Then, in 1994, Congress wrote peyote directly into federal law: an amendment to the American Indian Religious Freedom Act declared that the use of peyote by an Indian for bona fide traditional ceremonial purposes "is lawful, and shall not be prohibited by the United States or any State." Peyote became the one plant in Schedule I with an explicit federal right to use it. The protection has honest limits: it runs to members of federally recognized tribes, not to the general public, and when the Supreme Court held in 1997, in *City of Boerne v. Flores*, that the 1993 Act could not be imposed on the states, peyote's protection survived only because the 1994 statute rested on Congress's separate authority over Indian affairs. In 2006, when a small church sought federal permission to use ayahuasca, a sacramental tea brewed from two Amazonian plants, the Court in *Gonzales v. O Centro* pointed to the peyote exemption as proof that the government could accommodate such use. The people forced to lose first had become the precedent that freed others.

Beside the other plants in these chapters, one difference explains much of peyote's history. Peyote had no industry behind it. No company profited from it, no government taxed it, no lobby defended it. There was only a poor and marginalized religious community with little power. Alcohol

and tobacco, both addictive and far deadlier, stayed legal and were defended by the wealth they produced. Peyote, which harmed no one, stayed in Schedule I and had to be rescued twice by act of Congress, precisely because there was no money in protecting it. The absence of a paying constituency is itself part of why it was forbidden.

There is a bitter turn at the end. Just as the legal right to the sacrament was secured, the sacrament itself began to vanish. Wild peyote grows slowly and only within a small range, and habitat loss and overharvest have thinned it, so that the church declared a supply crisis in 1995 and botanists now list the species as declining. The right to the ceremony is safe on paper at last. Whether the plant will survive to supply it is another matter. The next chapter turns to a second Indigenous sacrament of vision, a brew of two Amazonian plants, whose oldest enemy was neither a federal agent nor a corporation but a church.

Chapter Six

Ayahuasca



The peyote meeting in the last chapter lasted all night around a fire, and so does this one, a continent to the south. In the western Amazon a healer whom the Spanish call a curandero boils the woody vine of *Banisteriopsis caapi* together with the leaves of a shrub called chacruna, and by the light of a small flame sings the songs called icaros while those who have come to be healed drink the bitter brew and wait. What follows is not recreation. It begins, for most, with hard vomiting, the purge that the tradition treats as part of the cleansing, and then hours of intense and often overwhelming vision. Ayahuasca is understood by its peoples as a medicine and a teacher, taken for healing, for divination, and for the mending of the spirit, and prepared for with a long and austere diet. Like peyote, it is a sacrament that the modern world files under the word "drug," and like peyote, its story is of a people punished for the way they reach the sacred.

The brew is a feat of Indigenous pharmacology so precise it is hard to credit as folk knowledge. The curanderos tell the story of the plant itself teaching them the combination. Whether that is true or a tale they keep for the gringos will never be known for certain, and the reality is likely lost in time, but the peoples of the Amazon have almost certainly known these plants for thousands of years. What is verified is the chemistry, a quirk of it unique to the region. The chacruna leaf contains dimethyltryptamine, or DMT, a powerful psychedelic that the human gut destroys on contact, so that swallowed by itself it does nothing at all. The caapi vine contains a different set of compounds that switch off the very enzyme that would destroy the DMT. Combine the two, as the Amazonian peoples learned to do, and the DMT survives to reach the brain. Two plants that do nothing of the kind alone, brewed together into a visionary medicine, by a discovery Western chemistry could not explain until the twentieth century. How old the practice is cannot be fixed exactly: a ritual bundle found in a Bolivian cave and dated to about a thousand years ago tested positive for both DMT and one of the vine's compounds, which shows a deliberate practice of the kind is old, though it does not prove the exact

modern recipe. And the brew is not harmless. Because the vine suppresses an enzyme the body also uses to break down many drugs and foods, ayahuasca can be dangerous, even fatal, taken alongside contraindicated pharmaceuticals; it strains the heart, can destabilize a vulnerable mind, and has killed people at unregulated retreats. It is not addictive in the way of the opioids, but not addictive is not the same as safe. Meeting your creator, or facing the most troubling recesses of your own mind, is not for the unprepared, and even the prepared meet disturbances that can leave a lasting wound. The danger is real when people ignore the protocols or the counsel of their curandero. That being said, to gauge the real harm ayahuasca does in the world, it would be tempting to use alcohol or tobacco as the reference, but the enormity of their harm would make the comparison too easy; and to weigh it against the many prescribed medicines that have supposedly passed the rigorous testing and randomized controlled trials our regulatory agencies call the gold standard for approval would be an unbalanced equation of another kind. A fairer comparison sits in any ordinary medicine cabinet: hundreds of people die every year from over-the-counter acetaminophen, and more die in traffic on the way to buy it. Ayahuasca's danger is modest, and it is not what got the plant forbidden.

For most of the plants in these pages the suppressor was a bureau or a corporation. Here, first, it was a church. When Spain conquered the Andes it set out to destroy the Indigenous religion it found, through a campaign its own priests named the extirpation of idolatry. Church inspectors toured the villages, gathered denunciations, and staged public burnings at which they destroyed sacred objects, pulled down shrines, dug up and burned the mummified ancestors, and flogged or exiled the native religious specialists. A priest named Francisco de Ávila prosecuted the old religion in the highlands and staged a public burning of sacred things in Lima's main square in 1609; a Jesuit named Pablo José de Arriaga wrote, in 1621, a working manual for how to do it. The Third Council of Lima ordered the destruction of khipu, the knotted cords in which Andean

peoples kept their records, and khipu were burned. The retelling often garbles what was destroyed: the Inca kept no books, only these cords, and it was the Maya, far to the north, whose written codices were thrown on a bonfire, by the Franciscan Diego de Landa at Maní in 1562, until only a handful survived. Direct colonial documentation of ayahuasca itself being outlawed is thin. But the broader record is clear: the colonial church set out to erase Indigenous visionary religion wherever it met it, treating a people's direct dealings with the spirit world as the devil's work to be stamped out.

Why a church would go to such lengths against a vision is not something the documents state outright, so what follows is an argument, not a settled fact. A church that stands between its people and the divine holds a kind of monopoly: it is the appointed road to God, and grace comes through its sacraments and its priests. A practice that offers a person direct, unmediated contact with what they take to be the source, with no priest and no church at all, is not merely a rival faith. It is a threat to the monopoly itself. And the hunger it feeds is old and worldwide. Peoples across history have sought transformation through a deliberate rehearsal of death and the dropping of the self: the vision quest of the North American plains, the death-and-rebirth initiations of the ancient Greek mysteries at Eleusis and around Dionysus, the egoic dissolution that ayahuasca and peyote reliably produce, which their drinkers experience as a communion with beings from beyond ordinary reality. This longing is universal, and it shows across the records. Whether those ancient mysteries were themselves induced alongside an entheogen or a hallucinogen is a separate and unsettled question. The ingredients of the secret rites were never written down, and the scholars tied to the idea, Wasson, Hofmann, and Ruck, and more recently Brian Muraresku, are still hunting the evidence, chiefly in the residues of ancient wine and beer. Only anecdote suggests it, the old tales of brews concocted by the priestesses of the age, unnamed plants stirred with eye of newt and wing of bat, which make the claim no more than a possibility. It has not

been proven as of this writing, though brews carrying the residue of ergot, a known hallucinogen that grows on moulding grain, have been reported in ancient vessels. The older point stands without that firm evidence: that the church suppressed a way of reaching the sacred which was not under its control is plain enough.



Fray Diego de Landa, who burned the Maya books at Maní in 1562.

Beside the corporate conduct documented elsewhere in these pages, the extirpation shows the same four moves, centuries early. Control the access: only the priest may mediate grace. Eliminate the rivals: extirpate the native healer and pull down the shrine. Define what is permitted and what is forbidden: the church's rite is a sacrament, the people's is idolatry and witchcraft. And manufacture the story that

justifies it: the chronicles that record the destruction call the old religion the falsehoods of the devil. The tobacco company and the drug maker guarded a market; the church guarded a monopoly on the soul. The method was the same.

Ayahuasca reached the United States mainly through two Brazilian churches, syncretic faiths that blend Amazonian practice with Christianity, the Santo Daime and the União do Vegetal, and it reached American law the way peyote had. When Customs seized a shipment of the tea bound for the small American branch of the União do Vegetal, the church sued under the Religious Freedom Restoration Act, the 1993 law Congress had passed to restore the protection that the peyote case, *Employment Division v. Smith*, had stripped away. In 2006, in *Gonzales v. O Centro*, the Supreme Court ruled for the church, unanimously, holding that the government had not shown a compelling reason to forbid its sacrament, and pointing directly to the government's own long-standing peyote exemption as proof that such an accommodation could work. The peyote of the last chapter had become the precedent that freed this one. A second case, in 2009, extended the same protection to the Santo Daime. But the limit of that protection is itself evidence of how unevenly the law is applied: it ran to organized churches able to fight a federal lawsuit, not to every Indigenous or independent seeker. Brazil, by contrast, simply authorized the sacrament, removing the vine from its prohibited list in the 1980s and reaffirming the right to religious use in 2010, while forbidding its sale as a commodity.

Then comes the last hypocrisy, the one that ties this plant to the rest. The same experience the church once punished, and the state still schedules, is now being patented and sold. As early as 1986 an American obtained a United States plant patent on a variety of the ayahuasca vine he had carried out of the Amazon; an Amazonian Indigenous coalition challenged it years later, and the patent was briefly rejected and then reinstated before it expired on its own, a case more tangled than the triumphant version usually told, but a real marker of the question of who may own a sacred plant.

Today the companies of the psychedelic revival hold patents on synthetic psilocybin down to the shape of its crystal; another firm carried MDMA-assisted therapy, which is not ayahuasca but belongs to the same commercial wave, to the edge of federal approval before regulators turned it back in 2024; and a global industry sells ayahuasca ceremonies to Western tourists while the Amazonian peoples who developed the knowledge are largely cut out of both the law and the money. The split runs through the law itself. DMT, the molecule, is produced naturally by the human body, where it has been detected in trace amounts and may be released in greater quantity as a person dies. It is banned worldwide all the same, yet the international drug-control board has stated that the plant preparations containing it are not internationally controlled, even as national courts treat the brew as contraband. The same drink is lawful or criminal depending on who holds it and where. The route to the transcendent is licensed for the powerful and patented by the corporation, and it remains a felony for the powerless, at least until they take their final breaths.

The people who worked out the pharmacology over centuries hold neither the exemptions, which went to churches, nor the patents, which went to firms. As with peyote, the plant whose original users had the least money behind them stayed forbidden to them the longest. The church guarded a monopoly on the soul. The next chapter turns to a third forbidden species of vision, a mushroom whose keepers the modern world did not so much forbid as expose.

Chapter Seven

Psilocybin



Peyote and ayahuasca were forbidden. The mushroom was, in a way, treated worse: it was found out. Its ruin began not with a law but with a magazine.

On a night in the summer of 1955, in the mountain town of Huautla de Jiménez in Oaxaca, a Mazatec healer named María Sabina led an all-night vigil of the kind her people had kept in secret for centuries. By the light of a candle she ate the small bitter mushrooms her language calls the holy children, and she chanted to find the source of a sickness. This was a velada, a ceremony of healing, and the mushrooms in it were medicine and sacrament, not recreation, exactly as the peyote of the tipi and the ayahuasca of the Amazon were. What made this night different from those chapters is what was done to the woman who led it, and it was not done by a bureau or a corporation but by the plain fact of Western attention.

Western culture is by now familiar with magic mushrooms. They are a group of fungi in the genus *Psilocybe*, of which there are some two hundred species. They range in size, form, and colour, and they work by an exact chemistry whose precision is part of the argument. The mushroom contains psilocybin, which does little on its own; the body strips it down to a second compound, psilocin, and it is psilocin that acts, on the same serotonin receptor through which the mescaline of peyote and the DMT of ayahuasca act. The three sacraments of these chapters are, at the level of chemistry, cousins. Like them, the mushroom is not addictive: it builds no physical dependence, produces no withdrawal, and its lethal dose is enormous next to the amount that has any effect. And like them, not addictive is not the same as safe. A frightening experience can drive a person to harm; the mushroom can precipitate lasting trouble in someone prone to psychosis; and its one reliably deadly danger is not the drug at all but the gathering of it, because a forager who mistakes a poisonous species for a *Psilocybe* can die of the self-inflicted error. To state the danger plainly is difficult, because the government records the harm from mushrooms far less precisely than it records death from pharmaceuticals.

The danger is real but collateral; it is modest by comparison; and regardless, it is, again, not what got the mushroom outlawed.

The mushroom was sacred in Mexico long before it was a felony. The peoples of the central highlands ate it in ritual for centuries before the Spanish came, and called it teonanácatl, a Nahuatl word usually rendered flesh of the gods; the Mazatec who would keep the rite alive called the mushrooms the little ones that spring forth, and the holy children. The Franciscan friar Bernardino de Sahagún, compiling his great record of Nahua life in the sixteenth century, described the ritual eating of the mushrooms at night, with honey, amid singing and weeping and visions. The colonial church treated this, as it treated the Andean and Amazonian rites of the last chapter, as devil-worship to be stamped out, and drove the ceremony underground. But it did not die. In the remote Sierra Mazateca it survived, in secret, for four hundred years, until it surfaced in the velada that María Sabina led.

The man she admitted to that ceremony was not a seeker but a banker. R. Gordon Wasson was a vice-president of J.P. Morgan and an amateur student of mushrooms, and by his own account he and his companion were the first outsiders allowed to eat the holy children in a Mazatec rite. Two years later, in May 1957, Wasson published what he had seen in *Life* magazine, in a long and admiring article titled "Seeking the Magic Mushroom," a phrase that would become iconic as a whole counterculture was born. The article reached millions, and it ruined her. Seekers poured into Huautla in the years that followed, the curious and the famous and the merely high, and María Sabina's own community turned on her for having opened the sacred vigil to outsiders. Her house was burned. She was harassed by the authorities the foreigners had drawn. And she came to believe the ceremony itself had been broken, that once outsiders took the mushrooms for reasons that had nothing to do with healing, the holy children lost their power. She died poor, having received almost none of the money or the fame that flowed from what she had shown the world. Here the question of

these pages takes its most personal form. Nothing was done to the mushroom. Everything was done to her, and the instrument was not a law but a magazine article and the flood of interest it generated.

Two things followed from that exposure, and together they set the pattern for all that came after. The first was that the mushroom became a molecule. Within a year of the article the Swiss chemist Albert Hofmann, who had made LSD, isolated and synthesized psilocybin at the pharmaceutical house Sandoz, which sold it for a time under a brand name. The plant belonged to the Mazatec; the compound went into a corporation's catalogue. The second was that the government took a diabolical interest, in secret. Under its now-exposed mind-control program MKUltra, the Central Intelligence Agency paid for Wasson's return expedition to Mexico in 1956, disguising the money as a grant from a medical foundation it used as a front, and arranged for one of its own chemists to travel with him and bring back samples. The limits of what can be said here matter. It is documented that the Agency funded the trip through that conduit and that it was interested in the drug; it is not documented that Wasson knew, or that the CIA had any hand in the *Life* article. The documented fact is enough, and it is its own kind of irony: the same government that would soon make the mushroom a felony had already paid, in secret, to carry it out of the mountains and into a laboratory.



Psilocybe aztecorum, a sacred mushroom of central Mexico.

Criminalization came with the decade that followed. The psychologist Timothy Leary and his colleague Richard Alpert ran a psilocybin project at Harvard until the university dismissed them in 1963, and the mushroom became bound up with a counterculture the country was learning to fear. Federal law reached the hallucinogens in 1965 and tightened in 1968, and in 1970 the Controlled Substances Act placed psilocybin in Schedule I, the tier of drugs with, in the law's own words, no accepted medical use and a high potential for abuse, alongside heroin; the international convention of 1971

did the same. All legal study of the mushroom and its derivatives was suspended. One note separates this plant from the last two. Peyote won its worshippers a legal exemption, and ayahuasca won one in court, but the mushroom's users got nothing of the kind. There was no church with standing to sue, and no exemption was ever written.

The ban did not stop the mushroom, and a book did more to spread it than any law did to contain it. In 1976 two brothers, Terence and Dennis McKenna, published under pen names a slim manual, *Psilocybin: Magic Mushroom Grower's Guide*, that laid out a simple way to grow the mushrooms at home from a spore print. Before it they were rare and obscure outside Oaxaca; after it, anyone with a jar and some grain could raise them, and it sold hundreds of thousands of copies. The brothers left two very different legacies. The practical one, a method to grow the fungus that no prohibition could reach, was real and lasting. The other was theory: Terence became the movement's spellbinding talker and spun large claims about the mushroom and the human mind, and his brother Dennis, who became a serious research pharmacologist, was among the first to doubt them.

And then, after thirty years in which the research sat frozen, the same molecule was reborn as a promising medicine, and the reversal is the sharpest illustration in these pages of how money moves the law. Beginning in 2006, researchers at Johns Hopkins and New York University published trials in which psilocybin, given under supervision, eased depression and the fear of dying, and the results drew serious attention. The Food and Drug Administration granted the drug its breakthrough-therapy status, first to a company called Compass Pathways in 2018 and then to a nonprofit institute in 2019, a designation that hurries a promising candidate through review but is not approval; psilocybin remains an experimental drug, not a licensed medicine. Compass, meanwhile, obtained patents on a synthetic crystalline form of psilocybin, down to the shape of the crystal, and when a nonprofit challenged those patents as claiming something

not truly new, the patent office declined to overturn them and they stood. States began to move on their own: Denver voted in 2019 to make the mushroom the lowest priority for its police, and Oregon in 2020 and Colorado in 2022 voted to build licensed programs where an adult could take psilocybin under supervision, for a fee. The whole of it is this. A molecule that is a felony in an ordinary person's pocket is at the same time patented by a corporation, hurried along by the federal drug regulator, and sold in licensed rooms to those who can pay, while the Mazatec who kept it alive, and María Sabina's heirs, hold neither the patents nor the profits.

There are many unanswered questions the little teachers do not explain. Terence McKenna, the same man who taught the country to grow the mushroom, proposed that eating it had driven the evolution of the human brain itself, an arresting idea with little evidence behind it and no way to test the hypothesis. McKenna also liked to say the only real danger of these drugs was death by astonishment, the sheer overwhelming force of the experience, a phrase the neuroscientist Andrew Gallimore later took for the title of a book that treats the visions as something to be explained rather than waved away. Others, the psychiatrist Rick Strassman writing of the related molecule DMT, and the writer Brian Muraresku on the ancient mysteries, argue that such compounds lie behind humanity's oldest religion. These are ideas that remain hypothetical, and for now untestable, like string theory, dark matter and dark energy, the paranormal, and the nature of consciousness itself. Science has trouble reaching such elusive things. Sometimes the mathematics points to a mystery, and consciousness is a phenomenon every human lives inside, but what these things finally are can be left, for now, to the domain of gods and mystics.

Around this fungus gather the largest questions in these pages, about what the experience is and where it came from. What is not unknown is the pattern, which points to a system that works to an end. Peyote, ayahuasca, and the mushroom were all sacraments whose original keepers had no industry,

no lobby, and no fortune to defend them, and that is exactly why they stayed forbidden to those keepers while others learned to profit from them. The next plant is the mirror image. It had all of that and more, a global industry, a marketing machine, and a family fortune built on selling it, and its story shows what becomes of a plant when money, and not a curandera, decides its fate.

Chapter Eight

Opium



The last three plants had no fortune behind the people who held them sacred. This one had fortunes to be made, and would become one of many sources of illicit funding for criminal organizations and covert agencies. But this is not a new phenomenon. It is a fact of our past, and opium comes with a very long history.

Opium is old almost beyond reckoning. The first written mention of the poppy appears on Sumerian tablets from the third millennium before Christ, and knowledge of it passed through Babylon, Assyria, and Egypt, whose physicians used it against pain. For most of human history it was the best medicine anyone had. Before ether and the surgeon's anaesthetic, which did not arrive until the middle of the nineteenth century, opium was very nearly the only thing that could dull severe pain, so that a patient facing the knife had otherwise little more than a strap or a stick to bite down on. It eased the dying, quieted the cough, stopped the killing diarrhea of dysentery, and brought sleep, and it did all of this while giving the user a warm, dreaming ease that people naturally sought again. In 1522 the Swiss physician Paracelsus dissolved opium in alcohol and named the result laudanum; in the 1680s the English doctor Thomas Sydenham refined his own famous version, and laudanum became the household remedy of the English-speaking world for the next two centuries. Then, around 1804, a young German apothecary named Friedrich Sertürner isolated the poppy's principal active compound, a pure alkaloid he named morphine after Morpheus, the god of dreams. For the first time a physician could deliver a measured, predictable dose of the poppy's power. Morphine was a genuine gift to medicine. It was also far stronger than the teas and syrups that came before it, and it left dependence in its wake wherever it went.



Mrs. Winslow's Soothing Syrup, 1876. Its principal ingredients were morphine and alcohol; it was sold for teething infants.

The plant's history also carries the clearest proof that the great pushers of opium were not the poor or the foreign but the most powerful empire of the age. Through the eighteenth and nineteenth centuries Britain wanted Chinese tea, silk, and porcelain, while China wanted little that Britain made, so British silver drained steadily east in payment. To reverse that flow, the British East India Company grew opium under monopoly in its Indian territories and sold it into China, against Chinese law, until by the 1830s the drug paid for the whole of the tea trade and had addicted a large part of the Chinese coast. When the imperial commissioner Lin Zexu moved to stop it in 1839, seizing and destroying the merchants' opium at Canton, Britain went to war to keep the trade open, and it won. The Treaty of Nanking that ended the First Opium War in 1842 forced China to pay a large indemnity, open five ports to foreign trade, and cede Hong Kong, and the opium kept flowing. What came next in America sharpens the irony: the Western powers that had forced opium on China at gunpoint would, within two generations, criminalize the Chinese for the very habit those powers had pressed upon them.

That criminalization began, in the United States, with a bottle on a nineteenth-century shelf. Mrs. Winslow's Soothing Syrup, sold from 1849 for fretful, teething babies, listed morphine and alcohol among its ingredients, roughly sixty-five milligrams of morphine to the ounce, so that a single teaspoon carried several times the dose an infant

could safely take. It quieted children, and it killed an unknown number of them; a later generation called it "the baby killer." The syrup was not an aberration. It was just one of many products that traded harm for profit. For most of the nineteenth century opium, the dried latex of the poppy *Papaver somniferum*, was respectable medicine in America too, carried in every doctor's bag and kept in most family cupboards, drunk as laudanum and sold across the counter for pain, cough, and sleeplessness, while the hypodermic needle and the Civil War spread injectable morphine and left dependence behind. When Britain moved to control opium in the Pharmacy Act of 1868, it classified the raw drug as dangerous but exempted the branded patent medicines that contained it. The plain drug was restricted; the corporate product stayed on the open shelf. From the beginning, money and corruption, not chemistry, decided who could get it.



A Bayer advertisement, 1899, listing Heroin among its pharmaceutical products.

Nothing about the poppy changed when the United States passed its first drug law. The user changed. In 1875 the San Francisco Board of Supervisors banned opium dens, forbidding anyone to keep, visit, or support a place where opium was smoked. It is generally counted as the first anti-drug law in the country, and its method of consumption somehow became the denominator that defined its legality. Smoking was the Chinese way of taking opium; laudanum, morphine, and the patent syrups were the white way. The

ordinance banned smoking and left morphine legal. The law's text never named the Chinese; it was facially neutral, forbidding a method rather than a people. But its purpose and its enforcement were not. The fear it answered was of white citizens, and especially white women, being corrupted inside Chinese establishments, and the dens the police raided were the ones that served whites. The ordinance sat within the anti-Chinese campaign that produced the Chinese Exclusion Act of 1882, the first federal law to bar a people by nationality, and when Congress banned the import of smoking opium in 1909 it did so less from any domestic emergency than to earn standing as China's ally against the drug. America's first drug prohibition was a race law, and like so many of the plants we have looked at, its prohibition wore the costume of public health.

The control went national in 1914. Two international meetings, at Shanghai in 1909 and The Hague in 1912, committed their signatories to confine opium to medical use, and the United States answered with the Harrison Narcotics Tax Act, signed by President Woodrow Wilson that December. Like the cannabis law that followed it, the Act was built as a tax: anyone dealing in opium or coca had to register and pay, and to handle the drug without registering became a crime. Its authors called it a revenue measure, but the Treasury and the courts read it further. In 1919, in *United States v. Doremus* and *Webb v. United States*, the Supreme Court allowed the government to use the Act to forbid physicians from prescribing opioids merely to maintain a patient who was already addicted. Being addicted was never itself written into the statute as a crime. But once a doctor could no longer prescribe to a dependent patient, that patient, cut off from any legal supply, was driven into the illicit market and made a criminal there. What had been a medical condition a physician could manage became, in practical effect, a police matter, and for the rest of the century opium was the illegal, foreign, criminal drug.

Then the same molecule was sold back to the country as a safe medicine, and the country believed it, because this time

a corporation was doing the selling. The groundwork was a single letter. In 1980 the New England Journal of Medicine published five sentences by Jane Porter and Dr. Hershel Jick, reporting that among nearly twelve thousand hospitalized patients given a narcotic, only four had become addicted. It was a brief note about short-term use, and everything turned on a detail its later users dropped: these had been hospital inpatients, given the drug briefly and under a nurse's watch, nothing like the millions who would later swallow a pill at home for months or years. Yet over the following decades the letter was cited hundreds of times as proof that opioids rarely addict. In 2017 the journal printed an unusual warning atop it, noting that it had been "heavily and uncritically cited." On that misread foundation, Purdue Pharma, owned by the Sackler family, launched the extended-release oxycodone pill OxyContin in 1996. That launch may be the most infamous development of a medicinal product in modern history. Purdue's sales representatives were trained to tell doctors that fewer than one percent of patients would grow addicted, a figure with no research behind it. Its signature claim, that a single dose lasted twelve hours, was the feature the drug was sold on, and Purdue knew the pill wore off sooner in many people; rather than have doctors prescribe it more often, which would have weakened the claim, the company urged them to prescribe a larger dose, which raised the risk of overdose. The molecule the country had spent a century treating as a criminal poison was now a blockbuster, because the hand that held it had changed. A corporation holds the legal status of a person, but a person bound by contract to its shareholders, one that behaves almost psychopathically in its founding rule that profit comes before harm.



A Bayer Heroin bottle.

The reckoning, when it came, fell the way it usually falls. In 2007 a Purdue corporate entity pleaded guilty in federal court to misbranding OxyContin as less addictive than the company knew it to be, and paid about six hundred million dollars; three of its executives pleaded guilty to a misdemeanor and received probation and community service, and no prison. As the lawsuits mounted, the Sacklers withdrew roughly ten billion dollars from the company, the larger part of its value, over the following decade. Purdue filed for bankruptcy in 2019, and in 2020 it pleaded guilty a

second time, to felony conspiracy, in a resolution the government valued at more than eight billion dollars; the family agreed to pay a separate civil sum and admitted no crime. A 2021 bankruptcy plan would have granted the Sacklers permanent immunity from opioid lawsuits in return for their contribution, until the Supreme Court struck the immunity down in 2024, in *Harrington v. Purdue Pharma*, Justice Neil Gorsuch writing that the bankruptcy law did not allow it. A revised settlement of about seven and a half billion dollars, this time with no immunity for the family, was approved in November 2025. The other end of the same plant's history ran the opposite way. The Chinese smoker of the 1870s faced the raid and, before long, the closed border; the modern street user faces prison. The company that misbranded a drug into a mass-casualty event paid money, and the family that owned it was never charged with a crime.

The toll is not in dispute, because the government keeps the count. Between 1999 and 2022 about seven hundred and twenty-seven thousand Americans died of opioid overdose, the annual number by the end roughly ten times what it had been at the start. The Centers for Disease Control and Prevention describes the dying in three waves: first the prescription pills, rising from the late 1990s; then heroin, as dependent people were cut off from the pills; then illicit fentanyl, a synthetic many times stronger than morphine, which now drives most of the deaths. The waves are a single pipeline, and the marketed pill primed it. The dead were concentrated where Purdue had marketed hardest, in rural and working-class and Appalachian communities, while the profit flowed upward to one of the wealthiest families in the country.



The destruction of seized opium at Canton, 1839.

The plant's supply has answered to power as plainly as its legality has, and nowhere more visibly than in Afghanistan, which for most of this century grew the majority of the world's illicit opium. In July 2000 the Taliban government banned poppy cultivation, and within a single season the crop nearly vanished, falling from about eighty-two thousand hectares in 2000 to roughly seven thousand in 2001, a collapse of some ninety percent that the United Nations Office on Drugs and Crime recorded at the time. That ban ended with the American-led invasion late in 2001. Across the twenty years of occupation that followed, cultivation climbed to record levels, and Afghanistan supplied the overwhelming share of the world's heroin while a Western military held the country. After the Americans withdrew and the Taliban returned, a second ban, in 2022, cut the crop by about ninety-five percent again, from two hundred thirty-three thousand hectares to under eleven thousand, by the same agency's count. The pattern itself is not in dispute: the poppy was suppressed before the occupation, flourished throughout it, and was suppressed again after it ended. What the pattern proves is narrower than the popular telling claims. The occupation coincided with the largest opium harvests in modern history; warlords and officials allied to

the occupation profited from the trade; and American counter-narcotics efforts failed for two decades. All of that is documented. That any United States agency deliberately organized the trade is not: the public record does not establish it. The documented fact is damning enough on its own: the same government that jailed its own citizens over these drugs presided over the greatest expansion of opium growing in modern history, and did not stop it.

The bottle of soothing syrup on the shelf comes back. One plant, held first by empires and profitable companies and left legal, then by a despised minority and outlawed, then by a corporation and sold at scale, and at every step the line between medicine and menace was drawn not by the poppy but by who profited from it and who paid. Opium shows the whole method in a single plant: criminalize it in the hands of the powerless, monetize it in the hands of the powerful, and manufacture the science that assures the public the profitable version is safe. That method is not unique to plants, and it neither began nor ended with the poppy. The chapters that follow leave the garden and pursue the method itself, into the industries that ran the same playbook with lead, with chemicals, and with the food on the family table.

PART THREE

The Playbook



Chapter Nine

The Corporate Playbook



In October 1924, in a refinery on the New Jersey shore, men began to go mad. They worked at the Standard Oil plant at Bayway, blending a new gasoline additive called tetraethyl lead, and one after another they fell into tremors, hallucinations, and violent delirium; some had to be held down in straitjackets. The workers took to calling the place the House of Butterflies, for the insects the poisoned men saw swarming the air. Five of them died at Bayway that month, and across the facilities that made the additive that year some fifteen workers died and dozens more were disabled. The companies that made it, General Motors, DuPont, and Standard Oil, already knew the compound was a potent poison. They also knew there was another way to keep engines from knocking, a blend using ordinary alcohol, which worked but could not be patented and sold at a profit. They chose the poison that could. And then, for the next sixty years, they kept the public from learning what they had known from the start.

That choice, and the system built to protect it, is the subject of this chapter, and the reason for it is clear. The argument of these pages is that a plant species' legal fate is set by money and power rather than by danger, and the natural objection is that this sounds like a conspiracy theory, that corporations do not really coordinate to bury the truth about their products. This chapter answers that objection with a reality that is uncomfortable for anyone who trusts their institutions. In case after case, in the records of courts and the documents pried out of company files, it has been proven that industries knew their products harmed people, paid for science to hide it, and delayed the reckoning for decades. The cases establish a method, not a conspiracy. The same sequence recurs: control the evidence, manufacture uncertainty, discredit inconvenient findings, and delay the reckoning until the profitable product has paid for itself. The lead and tobacco and chemical companies did not run the drug prohibitions. The playbook did. Turned in the opposite direction, the same machinery could make a competing plant appear uniquely dangerous: fund the studies that find the harm, elevate the findings that support prohibition, suppress

or dismiss the evidence that does not, and let the resulting certainty do the work of the law.

The story comes back to the beginning, to the lead and to the Standard Oil plant, which belonged to the Rockefeller family, whose fortune also built much of American medical education in the same era. When the deaths at Bayway forced a public alarm in 1925, the Surgeon General convened a committee, and in 1926 it reported that it found no good grounds to ban leaded gasoline so long as the workers were protected, while warning that the long study of the public's exposure still needed to be done. That study was never funded independently. It was handed, in effect, to the industry itself. For the next forty years the science of lead in America was dominated by one man, Robert Kehoe, chief medical adviser to the Ethyl Corporation and head of a laboratory the lead companies paid for. Kehoe's doctrine was that lead was natural in the body, that low doses did no harm, and that the burden of proof lay on anyone who claimed otherwise: show me the data. Because he controlled the funding, the data that would have disproved him was never allowed to exist. The spell broke in 1965, when a geochemist named Clair Patterson, who had built ultra-clean laboratories in order to measure lead at all, showed that modern humans carried something like a hundred times the lead of their pre-industrial ancestors, and that the rise tracked the burning of leaded gasoline. For saying so, Patterson found his research contracts pressured and his funding squeezed. Leaded gasoline was not phased out of American cars until 1986, and not off the earth until 2021, a century after the first men died making it. When it went, the proof went on the record with it: the share of American children with dangerous levels of lead in their blood fell from nearly nine in ten to about one in a hundred. The harm had been real, and preventable, for sixty years.

The doctrine Kehoe practiced on lead was named, and perfected, by the tobacco industry, and there it was finally proven in a court of law. The 1954 "Frank Statement to Cigarette Smokers" and the research body it founded were

built, as an earlier chapter showed, to keep a settled question looking open. What the industry's own files eventually revealed was a preconceived strategy built for deception. A 1969 memo from the Brown and Williamson company put it flatly: "Doubt is our product, since it is the best means of competing with the 'body of fact' that exists in the mind of the general public." And in 2006, in *United States v. Philip Morris*, Judge Gladys Kessler of the federal court in Washington found, in an opinion of more than sixteen hundred pages, that the companies had carried on a decades-long conspiracy to deceive the public, a fifty-year scheme to defraud. This is the fact that turns suspicion into proof: not a critic's charge, but a federal court's finding, that manufacturing doubt about a known harm, for profit, is exactly what was done.

Side by side, the cases resolve the method into a procedure, repeatable enough to number. First, fund the science yourself, so that the questions are yours to frame and the disproving study is never run, as Kehoe did for lead and, as an earlier chapter showed, a pharmaceutical executive's commission did for the coca leaf. Second, manufacture doubt: you need not prove the product safe, only make its harm look uncertain, which the tobacco companies raised to a creed, doubt is our product. Third, capture the referee, placing your own experts on the regulatory panels that judge you, as the lead industry did on the committee of 1926, and turning "more study is needed" into a permanent delay. Fourth, attack the messenger, as the lead interests did to Clair Patterson and the tobacco companies did to the researcher who exposed them, so that honesty costs an independent scientist their funding and their peace. Fifth, conceal: run the tests and keep the results in the file, as DuPont did with the poisoned tap water it measured in 1984. Sixth, delay, because every year of doubt is another year of sales, and sixty years passed between the first lead deaths and the ban. Seventh, when the lawsuits finally land, settle, often for a fraction of the profit and often under seal, as the tobacco companies did for two hundred billion dollars in

1998, admitting nothing, so the documents stay buried and the next community begins from zero.

None of this is history. The clearest modern proof runs through a family of industrial chemicals called PFAS, the "forever chemicals" that do not break down. From the 1950s the DuPont company used one of them, known as C8, to make Teflon at a plant on the Ohio River, and its own studies found the chemical toxic. In 1984 DuPont secretly tested the tap water of the town across the river, found its chemical in it, and told no one, not the town, not the water utility, not the regulators. It took the death of a farmer's cattle and a lawyer named Robert Bilott, who pried loose tens of thousands of internal documents, to force it into the open. A court-mandated science panel studied some seventy thousand exposed people and found the chemical probably linked to six diseases, among them kidney and testicular cancer. DuPont paid the Environmental Protection Agency a then-record penalty for having failed to report the risk, settled with the injured for hundreds of millions of dollars, and in 2023 the makers of these chemicals agreed to pay America's water utilities well over ten billion dollars to clean them up.

The settlement did not end the chemistry; it moved it. As the original compounds were banned, the companies replaced them with new ones, structurally altered, freshly patented, and sold as safe, and the cycle began again. When PFOA was phased out, a near relative the industry called GenX took its place, and within a few years it too was turning up in drinking water and drawing lawsuits and limits of its own. Regulators have a name for the maneuver, regrettable substitution: a banned molecule is nudged into a shape the old rules do not yet cover, and the reckoning restarts from zero. It is the seventh step run again: settle, admit nothing, return to the laboratory, and profit for as long as the new molecule holds, then repeat.

The chemicals are now in the blood of almost every person in the country. The machine that sold leaded gasoline was still running, on schedule, into our own decade.

If this were an isolated event it would strain belief on its own, but there are many to choose from, and two of them close the door on disbelief from opposite sides. Thalidomide was a sedative sold from the late 1950s across Europe, Britain, Canada, and dozens of other countries, and given to pregnant women for morning sickness. It caused an estimated ten thousand children to be born with malformed or missing limbs, and thousands more to die before birth. It never reached the American market, because a single reviewer at the Food and Drug Administration, Frances Kelsey, refused to approve it, holding out against the manufacturer's pressure because its safety data was thin. The catastrophe that unfolded across the rest of the world was the measure of what her refusal spared the United States, and it moved Congress in 1962 to require, for the first time, that a drug be proven both safe and effective before it could be sold. An uncaptured regulator can stop the harm.

The other case shows what happens when no one refuses. From 1932 to 1972 the United States Public Health Service, an arm of the federal government, ran a study of untreated syphilis in and around Tuskegee, Alabama, enrolling some six hundred poor Black men, about four hundred of whom had the disease. The men were never told they had syphilis; they were told they were being treated for "bad blood," and they were watched, without treatment, so that government doctors could observe the illness run its course to the grave. Penicillin became the standard cure by the late 1940s, and it was deliberately withheld from them for another quarter of a century. This was not the work of a rogue. It was designed, funded, and published by a federal health agency, carried on in plain sight in the medical literature, and it ended only in 1972, when one of its own investigators, Peter Buxtun, took it to the press. The outrage forced the country to write the modern rules of informed consent. An institution will conceal a lethal harm for decades if no one makes it stop. The machine can be beaten, and it will run until it is.

Held against the plants in these chapters, the whole procedure fits, aimed the other way. Everything the reader

has just watched keep a poison legal is the same mechanism used to make a plant illegal. To protect a profitable harm, manufacture doubt about its danger; to eliminate an unprofitable plant, manufacture certainty of it. Fund the studies that find what you need, capture the body that schedules the drug, discredit the researcher who reports a benefit, bury the data that exonerates the plant, and delay the reckoning for fifty years. The playbook is symmetric. It manufactured doubt to keep leaded gasoline in the tank, and it manufactured danger to keep the coca leaf out of the Andean farmer's hand. And its most intimate application, the one that reaches the family table three times a day, was never a fuel or a chemical or a drug at all. It was food.

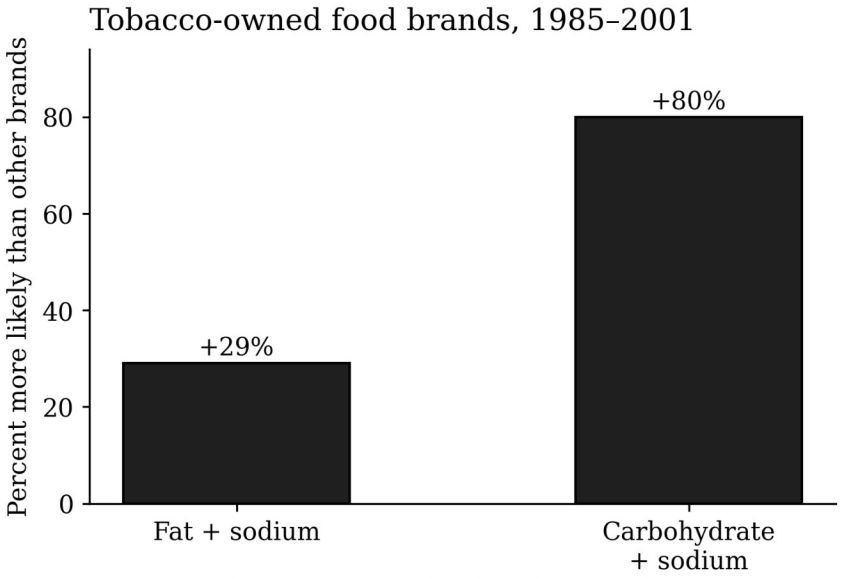
Chapter Ten

Engineered Food Addiction



The product began as a problem: too much bologna. In the early 1980s the Oscar Mayer company had a surplus of processed meat that was losing its appeal, and a team led by a manager named Bob Drane was asked to find a way to sell more of it. What they designed, in 1985, was a small tray of crackers, meat, and cheese aimed squarely at children and the harried parents who packed their lunches. They called it Lunchables. It was heavy with salt and saturated fat, built for convenience and for the young palate, and it worked: two hundred million dollars in sales its first year, half a billion within a decade. The bologna problem was solved, not by making the meat more nourishing, but by engineering it into something a child would crave. That is the move this chapter follows, because the company that owned Oscar Mayer when Lunchables launched was not, in the end, a food company at all. It was Philip Morris, the largest cigarette maker in the world.

The overlap was not the accident of a single brand. For the better part of two decades the American pantry belonged to the tobacco industry. Philip Morris bought General Foods in 1985 for around five and a half billion dollars, and Kraft in 1988 for some thirteen billion, and merged them into the largest food company in the country, the maker of Maxwell House, Jell-O, Kool-Aid, Oscar Mayer, and the rest. In the same years R. J. Reynolds joined with Nabisco. From roughly 1985 to 2001, the two largest food companies in the United States were owned by the two largest tobacco companies, and it was in exactly those years that the modern engineering of craving into food was scaled across the supermarket. This is not an inference. It is the corporate record.



Tobacco-owned food brands were more likely than others to meet the hyperpalatable threshold, 1985–2001.

What those companies carried into food was a discipline they had learned on the cigarette. The tobacco chemists had tuned nicotine's delivery to the exact threshold of the brain's reward; the food scientists learned to do the same with fat, sugar, and salt. Howard Moskowitz, the food scientist profiled in Michael Moss's *Salt Sugar Fat*, taught the industry to find what he called the bliss point, the precise balance of sweetness or richness at which craving peaks and fullness is held off, running a product through dozens of formulations until the tongue could not refuse it. David Kessler, the former head of the Food and Drug Administration who had fought to regulate nicotine as an addictive drug, later described the result as conditioned hypereating, food built to trip the same circuits that drive any compulsion. And the optimization is not confined to a few snacks. When the researcher Tera Fazzino and her colleagues built the first numerical definition of a hyperpalatable food, they found that around three in five of the products in the American food supply met it.

The link between the tobacco owners and the craving-tuned food is not only a matter of timing. In 2023 Fazzino and her colleagues went into the same archive of tobacco-company documents that the cancer lawsuits had forced open, established which food brands the tobacco firms had owned, and measured them against the rest of the food supply over time. During the years of tobacco ownership, those brands were significantly more likely than others to be built to the hyperpalatable threshold, the balance of fat, sugar, and salt at which craving peaks. Fazzino's team measured this two ways. By the combination of fat and sodium, the tobacco-owned brands were twenty-nine percent more likely to qualify; by the combination of carbohydrate and sodium, eighty percent more likely. And in 2022 the psychologist Ashley Gearhardt and her colleagues took the very criteria the 1988 Surgeon General's report had used to classify cigarettes as addictive, applied them to ultra-processed food, and found that it cleared the bar. What the record supports is a pattern, not a confession: that ownership predicted craving-optimized formulation, and that the products behave, by the government's own tobacco test, like an addictive substance. Whether anyone sat in a room and set out to addict the country is not something the documents prove, and this chapter does not claim it. The pattern is enough to show what the purpose of the system was.



Coca-Cola advertisement, c. 1900. The coca leaf gave the drink its name; once the cocaine was removed, what remained and was engineered ever upward was the sugar.

The defense of the food, once the science caught up, ran through the seven steps of the last chapter almost exactly. The scholars Kelly Brownell and Kenneth Warner laid the parallel out in 2009: the food industry, like the tobacco industry before it, funded its own science, stood up front groups that spoke as though independent, cast doubt on the studies it disliked, marketed to children, and lobbied the bodies that write the government's dietary advice. The

physician Casey Means, in *Good Energy*, gathers the figures: food companies outspend the National Institutes of Health many times over on nutrition research; industry-funded studies of sugary drinks return the industry's preferred answer far more often than independent studies do; and the great majority of the experts on a recent federal dietary-guidelines panel carried financial ties to the food or drug industry. Kim Nguyen and Stanton Glantz documented the most direct transfer of all: Philip Morris moved its machinery for marketing to minority communities straight from its tobacco division into its food division. The same firm, the same experts, the same playbook, pointed now at the palate.

The cost of this fell where the cost in these chapters always falls. The refined inputs of the modern processed diet are plants, stripped to their profitable fractions: corn made into syrup and starch and oil, cane and beet into sugar, soy and oil palm into cheap fat. American farm policy has for decades subsidized exactly these commodity crops while offering the growers of fruits and vegetables almost nothing, so that the engineered food is made artificially cheap and the fresh food is not. And the diseases that follow, obesity and diabetes and their long train, gather hardest in the poor and minority neighborhoods that the industry's own marketing was aimed at, and where fresh food is farthest away. Here the plant is not forbidden. It is conscripted. The same machinery of money and power that made the coca leaf a felony in an Andean cheek turned corn and sugar into an engine of preventable sickness in an American child, and called it lunch. What that sickness costs, and who profits from treating it rather than preventing it, is the question the last part of this book must answer.

Chapter Eleven

The Purpose of a System Is What It Does



The chapters before this one made an argument. This one makes none. What follows is a record, arranged by year, of cases in which a company or a government was held to account, in a court or by a regulator, for harm done to people or to the living world. Each entry is only the facts: what the product or the project was, what the harm was, and what the record found. No motive is assigned here, and no conclusion is drawn. The reader is left to do with the record what the record invites.

The chapters before this one traced how a handful of plants were made into crimes or into fortunes, and named the method by which it was done: fund the science, manufacture doubt or danger, capture the referee, and delay the reckoning until the profitable harm has paid for itself. The purpose of this chapter is to show that the method was never confined to those plants, or to the drug laws, or to any one industry. The same pattern runs through the whole economy, and it has been proven, over and over, in the one place built to test it, a court of law. What follows is that proof, gathered in a single place so the pattern can be seen at its full length.

The cases told in full here were chosen from a documented record of more than four hundred, and they were chosen by a single test: that the harm is clear and verifiable, established by a verdict, a criminal plea, a settlement, a regulatory finding, or a negotiated concession. The appendix that closes the chapter lists more than a hundred further cases by year and name alone, and even the two lists together are only a portion of what the courts and the regulators have set down. They are the ones that cost lives, or health across a lifetime, or a river, a forest, and the creatures that depended on it. They are, in one place, much of the worst that industry has done to the people and the places around it, and been made to answer for.

They are also only the ones that were found out, sued, and beaten. That is worth holding in mind while reading, because the population in whose name most of these harms were produced was, for the most part, not marching in the streets

to stop them. It was buying the car, the cigarette, the soft drink, and the cheaper gadget, and asking few questions about what the making of those things did downstream. Rivers were filled with what the factories poured out and with the plastics the convenience left behind, the arteries of the earth silting up with the same patience that saturated fat and plaque lay down in the arteries of the people who bought the products. The record below is the account of that exchange as the courts finally wrote it.

A note on how to read the entries. The outcome types are not equal, and the difference matters. A **verdict** means a court or jury found the company liable. A **criminal** outcome means a conviction or a guilty plea. A **regulatory** outcome means a government agency made a formal finding or imposed a penalty. A **concession** means a non-monetary result: a recall, an injunction, a protected area, a mandated program. And a **settlement** means money was paid to end the case, very often, by the company's own insistence, with no admission of liability at all. Where that is so, it is said. A settlement without admission is not a finding of guilt. It is a payment made to make a case go away, which the reader may weigh as they see fit.

The record runs from 1958 to the present, and it does not thin out as it approaches our own time.

The 1950s and 1960s

1958. Cutter Laboratories, Salk polio vaccine. A production run of the new polio vaccine contained live virus and paralyzed children who received it. *Verdict:* a California jury found the company not negligent but liable for breach of implied warranty, and the judgment was affirmed on appeal. (*Gottsdanker v. Cutter Laboratories*, Cal. Ct. App., 1960.)

The 1970s

1970. Grünenthal, thalidomide (Contergan), West Germany. The sedative, sold to pregnant women for morning sickness, was associated with severe birth defects in

their children. *Criminal proceedings and settlement:* the criminal case against company executives was terminated in 1970 without a conviction; in a parallel resolution the company paid 100 million Deutsche Marks into a fund for affected children, and the government added a like sum. (Aachen proceedings, 1968 to 1970; Contergan settlement, 1970.)

1973. Chisso Corporation, Minamata, Japan. Mercury-laden factory wastewater discharged into Minamata Bay poisoned the people who ate fish from it, causing the neurological illness that took the bay's name. *Verdict:* the Kumamoto District Court found the company negligent and ordered compensation. (Kumamoto District Court, March 1973.)

1975. Government of Quebec and Hydro-Québec, James Bay. [Indigenous] Large-scale hydroelectric development flooded and altered Cree and Inuit lands. *Concession:* the James Bay and Northern Quebec Agreement recognized defined Cree and Inuit rights and provided CA$225 million in compensation over roughly twenty years in exchange for permitting the development. (James Bay and Northern Quebec Agreement, 1975.)

The 1980s

1980. Diethylstilbestrol (DES) manufacturers, United States. Daughters of women given the anti-miscarriage drug developed cancers but could not prove which maker had supplied their mothers' pills. *Verdict and doctrine:* the California Supreme Court adopted market-share liability, apportioning damages among the makers by their share of the market. (*Sindell v. Abbott Laboratories*, Cal., 1980.)

1981. Ford Motor Company, Ford Pinto. A rear-impact collision ruptured the fuel tank and caused fatal and disfiguring burns. *Verdict:* a California jury awarded compensatory and punitive damages, the punitive sum reduced on the trial court's order, and the appellate court

affirmed, citing evidence that the company had gone to production knowing of the defect. (*Grimshaw v. Ford Motor Co.*, Cal. Ct. App., 1981.)

1984. Dow, Monsanto, Diamond Shamrock and other manufacturers, Agent Orange. The dioxin-bearing defoliant supplied for use in Vietnam was alleged to have harmed exposed U.S. veterans. *Settlement, without admission of liability:* $180 million paid into a class fund on the eve of trial. (*In re Agent Orange Product Liability Litigation*, E.D.N.Y., 1984.)

1985. Reed Inc. and Great Lakes Forest Products, with Ontario and Canada. [Indigenous] Mercury from a Dryden, Ontario paper and chlor-alkali operation was discharged into the Wabigoon and English River system, poisoning the Grassy Narrows and Wabaseemoong First Nations. *Settlement:* a 1985 agreement paid roughly CA$16.7 million and created a Mercury Disability Board and Fund for affected members. (Grassy Narrows and Islington Bands Mercury Disability agreement, 1985.)

1989. A.H. Robins, Dalkon Shield IUD. The contraceptive device was associated with pelvic infection, infertility, and deaths, drawing more than 300,000 claims. *Settlement, by bankruptcy plan:* the company's confirmed plan funded a claimants' trust of about $2.475 billion, which paid claims until it closed in 2000. (*In re A.H. Robins Co.*, U.S. Bankruptcy Court, E.D. Va.)

1989. Union Carbide Corporation, Bhopal, India. The 1984 release of methyl isocyanate gas from the Bhopal plant killed and injured thousands. *Settlement:* $470 million, approved by the Supreme Court of India in 1989. (*Union Carbide Corporation v. Union of India*, 1989.)

The 1990s

1992. Centre National de Transfusion Sanguine, France, HIV-contaminated blood products. Hemophiliacs were transfused with blood-clotting products

the transfusion service kept distributing after the contamination risk was known. *Criminal:* the Paris court convicted the service's former director, Michel Garretta, of fraud over product quality and sentenced him to four years in prison, with the former research director also convicted; the convictions were largely upheld on appeal in 1993. (Tribunal de grande instance de Paris, 1992; Paris Court of Appeal, 1993.)

1994. Exxon, Exxon Valdez. [Indigenous, in part] The 1989 grounding of the tanker spilled some eleven million gallons of crude into Prince William Sound, Alaska, harming fisheries and Native subsistence. *Verdict:* a federal jury awarded $5 billion in punitive damages, cut to $2.5 billion on appeal and to about $507.5 million by the U.S. Supreme Court in 2008. (*Exxon Shipping Co. v. Baker*, 2008.)

1994. McDonald's Restaurants, hot coffee. Coffee served at roughly 180 to 190 degrees caused third-degree burns requiring skin grafts. *Verdict:* a New Mexico jury awarded compensatory and punitive damages, the punitive sum reduced by the judge, after which the parties settled confidentially. (*Liebeck v. McDonald's Restaurants*, N.M. Dist. Ct., 1994.)

1995. Occidental Chemical and Hooker Chemical, Love Canal, New York. Chemical waste buried in the 1940s and 1950s later contaminated a residential neighborhood built over it. *Settlement:* $129 million to reimburse federal cleanup costs, ending sixteen years of litigation. (U.S. Department of Justice, 1995.)

1995. The Crown, New Zealand, Waikato-Tainui. [Indigenous] The 1860s invasion of the Waikato and the confiscation of more than 1.2 million acres. *Concession:* the Waikato Raupatu Claims Settlement Act delivered a formal Crown apology and a redress package valued at NZ$170 million, including returned land. (Waikato Raupatu Claims Settlement Act, 1995.)

1996. BHP, Ok Tedi mine, Papua New Guinea. [Indigenous] Riverine disposal of mine tailings degraded the Ok Tedi and Fly River systems on which downstream communities lived. *Settlement and concession:* an out-of-court settlement provided compensation and commitments toward tailings containment; the cash figures vary by source and are disputed. (*Dagi v. BHP*, Supreme Court of Victoria, 1996.)

1996. Pacific Gas and Electric, Hinkley, California. Hexavalent chromium used as a rust inhibitor migrated into the town's groundwater. *Settlement:* $333 million to more than 600 plaintiffs, then the largest such class settlement in U.S. history, resolving the claims without a trial verdict. (Hinkley groundwater litigation, San Bernardino County, 1996.)

1996 to 2000. Green Cross Corporation, HIV-tainted blood products, Japan. The firm kept selling unheated blood-clotting products after a heat-treated alternative existed, and hemophiliac patients were infected with HIV. *Criminal:* three former company presidents were convicted of professional negligence resulting in death. (Osaka District Court, 2000.)

1998. The major cigarette manufacturers, Tobacco Master Settlement Agreement. The states sought to recover the Medicaid cost of treating smoking-related illness. *Settlement, without admission of liability:* the four largest makers agreed with 46 states and other jurisdictions to pay at least $206 billion over the first 25 years, with payments continuing beyond, and to accept marketing restrictions. (National Association of Attorneys General, 1998.)

1999. MacMillan Bloedel, Clayoquot Sound, British Columbia. [Indigenous] Old-growth logging of the Clayoquot valleys, opposed in the mass blockades of 1993. *Concession:* a 1993 land-use decision protected part of the area, and the company later folded its Clayoquot logging rights into a joint venture with the Nuu-chah-nulth, Iisaak Forest Resources. This was a negotiated and administrative

outcome, not a court verdict. (B.C. land-use decision, 1993; Iisaak venture, circa 1998 to 1999.)

The 2000s

2000. Cigarette manufacturers, Engle, Florida. A class of Florida smokers alleged smoking-caused disease. *Verdict, later vacated:* a jury returned a $145 billion punitive award, which was vacated and the class decertified on appeal; the jury's underlying findings were preserved and seeded thousands of individual "Engle progeny" suits. (*Engle v. Liggett Group*, Fla., 2006.)

2000 to 2001. Bridgestone/Firestone and Ford, ATX and Wilderness tires. Tread separation on certain tires was linked to rollover crashes. *Concession:* a recall of some 6.5 million tires and the federal TREAD Act that followed; the death toll commonly cited is a figure reported by the National Highway Traffic Safety Administration, not a single court finding. (NHTSA recall, 2000.)

2001. Mayagna (Sumo) Awas Tingni Community v. Nicaragua. [Indigenous] The state failed to demarcate and protect the community's customary land while granting a logging concession over it. *Concession, by judgment:* the Inter-American Court of Human Rights ordered Nicaragua to demarcate and title the land, and title to about 73,000 hectares was issued in 2008. (IACtHR, 2001.)

2001. TAP Pharmaceutical Products, Lupron (leuprolide). The company ran a fraudulent pricing and kickback scheme around the prostate-cancer drug, inducing providers to bill government programs for samples given to them free. *Criminal:* TAP pleaded guilty to a conspiracy to violate the Prescription Drug Marketing Act and paid $875 million, comprising a $290 million criminal fine and roughly $559.5 million in civil False Claims Act payments. (U.S. Department of Justice, 2001.)

2002. Government of Quebec and Hydro-Québec, Paix des Braves. [Indigenous] Further hydroelectric

development on Cree land and the implementation of the 1975 obligations. *Concession:* a negotiated agreement providing recognition and long-term payments in exchange for permitting development. (Paix des Braves, 2002.)

2003. Monsanto and Solutia, Anniston, Alabama. Decades of PCB manufacturing waste contaminated the Anniston community. *Settlement, without admission of liability:* a $700 million global settlement, including a fund for roughly 21,000 plaintiffs. (Solutia SEC filing, 2003.)

2004. Warner-Lambert (Pfizer), Neurontin (gabapentin). The Parke-Davis division promoted the epilepsy drug for a range of unapproved uses, including bipolar disorder, pain, and migraine. *Criminal:* Warner-Lambert pleaded guilty to two felony counts of misbranding and paid more than $430 million, including a $240 million criminal fine, to resolve criminal and civil liability. (U.S. Department of Justice, 2004.)

2006. Government of Canada, Indian Residential Schools Settlement Agreement. [Indigenous] The residential-school system's removal and abuse of Indigenous children. *Settlement:* the largest class settlement in Canadian history, providing a Common Experience Payment of about CA$1.9 billion, an Independent Assessment Process for serious-abuse claims, and funding for the Truth and Reconciliation Commission. (IRSSA, approved 2006.)

2006. Philip Morris USA and other manufacturers, United States v. Philip Morris. A decades-long effort to deny the health effects and addictiveness of smoking, and to deny marketing to the young. *Verdict:* Judge Gladys Kessler held the companies civilly liable under the federal RICO statute and imposed injunctive remedies, including corrective statements. No monetary damages were awarded; this was a liability finding. (*United States v. Philip Morris USA*, D.D.C., 2006.)

2007. The Purdue Frederick Company, OxyContin. The opioid was misbranded as less addictive and less prone

to abuse than the record supported. *Criminal:* the company pleaded guilty to felony misbranding and three executives to misdemeanor misbranding, with total payments of about $634.5 million. (U.S. Attorney, W.D. Va., 2007.)

2007. State of South Australia, Trevorrow. [Indigenous] The unlawful removal of Bruce Trevorrow as an infant from his Aboriginal family. *Verdict:* the Supreme Court of South Australia found the state liable and awarded damages, the first successful Stolen Generations damages judgment. (*Trevorrow v State of South Australia*, 2007.)

2007. Merck, Vioxx (rofecoxib). The withdrawn painkiller was linked in litigation to heart attacks and strokes. *Settlement, without admission of liability:* $4.85 billion to resolve roughly 27,000 personal-injury claims. In 2011 Merck separately pleaded guilty to a misdemeanor misbranding charge and resolved civil claims for $950 million. (Master Settlement Agreement, 2007; U.S. DOJ, 2011.)

2009. Pfizer and Pharmacia & Upjohn, Bextra and other drugs. Off-label promotion of the anti-inflammatory Bextra and related conduct. *Criminal and civil:* a subsidiary pleaded guilty to a felony misbranding charge, and the total resolution was $2.3 billion, then the largest health-care-fraud settlement in Department of Justice history. (U.S. DOJ, 2009.)

2009. Eli Lilly, Zyprexa. Off-label promotion of the antipsychotic to elderly dementia patients. *Criminal and civil:* a misdemeanor misbranding plea and a total resolution of about $1.415 billion. (U.S. DOJ, 2009.)

2009. Pfizer, Trovan trial, Kano, Nigeria. An unapproved-drug trial conducted in children during a meningitis epidemic. *Settlement, without admission:* about $75 million. (Out-of-court settlement, 2009.)

2009. W.R. Grace, Libby, Montana. Asbestos-contaminated vermiculite from the company's mine was linked to widespread disease and death in the town.

Regulatory: the Environmental Protection Agency declared a Public Health Emergency at Libby, the first ever under Superfund law. The parallel federal criminal case ended in acquittal. (EPA, 2009.)

2009. ASARCO. Smelting and mining contamination at more than eighty sites. *Settlement, by bankruptcy:* $1.79 billion for environmental cleanup, then the largest such recovery in the country. (U.S. DOJ and EPA, 2009.)

2009. Royal Dutch Shell, Wiwa v. Royal Dutch Petroleum. [Indigenous] Alleged complicity with Nigerian security forces in abuses against Ogoni leaders in the Niger Delta, including the 1995 executions of Ken Saro-Wiwa and others. *Settlement, without admission of liability:* $15.5 million, part of it funding a trust for the Ogoni people. (S.D.N.Y., 2009.)

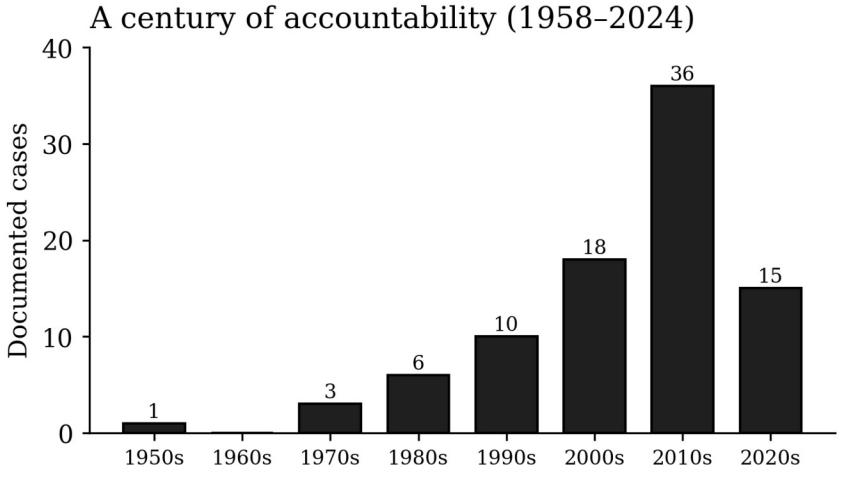
The 2010s

2010. Endorois Welfare Council v. Kenya. [Indigenous] Eviction of the Endorois people from their land at Lake Bogoria to create a game reserve. *Finding:* the African Commission on Human and Peoples' Rights found the eviction a violation and recommended restitution, a finding Kenya has largely not implemented. (African Commission, 2010.)

2010. AstraZeneca, Seroquel (quetiapine). The company marketed the antipsychotic for uses the Food and Drug Administration had not approved, including in the elderly and for anxiety and sleeplessness. *Settlement, without admission of liability:* $520 million to resolve federal and state civil False Claims Act allegations. (U.S. Department of Justice, 2010.)

2010. Trafigura, Probo Koala toxic waste, Netherlands and Ivory Coast. Caustic petroleum slops refused for treatment in Amsterdam were shipped to Abidjan and dumped, and the company concealed the waste's harmful nature. *Criminal:* the Amsterdam District Court

convicted Trafigura of illegally exporting hazardous waste and concealing its nature and imposed a fine of about 1 million euros; the firm had separately paid roughly 157 million euros to Ivory Coast and about 40 million euros to settle a UK victims' claim. (Amsterdam District Court, 2010.)



Documented corporate-accountability cases by decade, 1958–2024.

2011. Wyeth, DTP vaccine. Parents brought a design-defect claim after a child's seizures following vaccination. *Verdict for the manufacturer:* the U.S. Supreme Court held that the 1986 vaccine-injury statute preempts such design-defect suits. Included here as the legal keystone of that statute; it is a ruling on preemption, not a finding of harm. (*Bruesewitz v. Wyeth*, 2011.)

2011. United States, Cobell v. Salazar. [Indigenous] The federal government's mismanagement of individual American Indian trust accounts and lands. *Settlement:* $3.4 billion, roughly $1.5 billion in direct payments and $1.9 billion to buy back and consolidate fractionated trust land. (*Cobell v. Salazar*, D.D.C., final approval 2011.)

2011. Chevron, Aguinda / Lago Agrio, Ecuador. [Indigenous; contested] Amazonian oil contamination attributed to Texaco's operations from 1964 to 1990. *Contested verdict:* an Ecuadorian court entered a judgment

against Chevron, reduced to about $9.5 billion; a U.S. federal court then found the judgment had been procured by fraud and barred its enforcement, and courts in other countries also declined to enforce it. The judgment remains unpaid. Recorded here with both facts stated. (Ecuadorian judgment, 2011; *Chevron v. Donziger*, S.D.N.Y., 2014.)

2012. GlaxoSmithKline, Paxil, Wellbutrin, Avandia. Unlawful promotion of drugs for unapproved uses and failure to report safety data. *Criminal and civil:* GSK pleaded guilty and paid $3 billion, then the largest health-care-fraud settlement in U.S. history. (U.S. DOJ, 2012.)

2012. Abbott Laboratories, Depakote. Off-label promotion of the drug to dementia patients. *Criminal and civil:* a misdemeanor plea and a $1.5 billion resolution. (U.S. DOJ, 2012.)

2012. Amgen, Aranesp (darbepoetin alfa). The company sold the anemia drug for off-label doses the Food and Drug Administration had rejected and for an off-label use it had never approved. *Criminal:* Amgen pleaded guilty to illegally introducing a misbranded drug into interstate commerce and paid $762 million, including a $136 million criminal fine, $14 million in forfeiture, and $612 million in civil settlements. (U.S. Department of Justice, E.D.N.Y., 2012.)

2012 to 2013. BP, Deepwater Horizon. The 2010 blowout of the Macondo well killed eleven workers and spilled crude across the Gulf of Mexico. *Criminal:* BP pleaded guilty in 2012 to fourteen counts, including eleven of felony manslaughter, and was sentenced in January 2013 to about $4 billion in criminal fines and penalties. (U.S. DOJ; guilty plea 2012, sentence January 2013.)

2012. Kichwa People of Sarayaku v. Ecuador. [Indigenous] Oil exploration authorized on Sarayaku territory, with explosives placed there, without consultation. *Concession, by judgment:* the Inter-American Court found

Ecuador in violation and ordered remediation, consultation guarantees, and compensation. (IACtHR, 2012.)

2013. Johnson & Johnson and Janssen, Risperdal and Invega. Promotion of the antipsychotic for unapproved uses, including in elderly dementia patients and children, with kickbacks. *Criminal and civil:* a misdemeanor plea and a resolution of more than $2.2 billion. (U.S. DOJ, 2013.)

2013. DePuy and Johnson & Johnson, ASR metal-on-metal hip. The implant failed and shed metal debris into patients. *Settlement:* a compensation program valued at about $2.5 billion. (Settlement program, 2013.)

2013. Ranbaxy USA, adulterated generic drugs. The generic manufacturer distributed adulterated drugs made at its Indian plants and made false statements to the Food and Drug Administration about its manufacturing. *Criminal:* Ranbaxy pleaded guilty to seven felony counts and paid $500 million, comprising $150 million in criminal fine and forfeiture and $350 million in civil settlements. (U.S. Department of Justice, 2013.)

2013 to 2016. Poly Implant Prothèse, breast implants, France. The company filled implants with industrial-grade rather than medical-grade silicone. *Criminal:* the founder, Jean-Claude Mas, was convicted of aggravated fraud, upheld in 2016. (French courts, 2013 to 2016.)

2014. Toyota Motor Corporation. The company made deceptive statements to regulators and the public about defects causing unintended acceleration. *Concession, by deferred prosecution:* a $1.2 billion penalty and an independent monitor, the charge deferred rather than a conviction entered. (U.S. DOJ, S.D.N.Y., 2014.)

2014. Boehringer Ingelheim, Pradaxa (dabigatran). Plaintiffs alleged the blood thinner's bleeding risks, with no reversal agent then available, were not adequately warned of. *Settlement, without admission of liability:* $650 million to

resolve roughly 4,000 U.S. claims. (Boehringer Ingelheim, 2014.)

2014 to 2015. Anadarko and Kerr-McGee, Tronox. Legacy contamination from decades of chemical, mining, and oil-and-gas operations. *Settlement:* $5.15 billion, most of it for remediation, then the largest environmental enforcement recovery in Department of Justice history. (S.D.N.Y., 2014.)

2015. General Motors, ignition switch. A concealed defect could cut power and disable airbags; the company acknowledged fifteen related deaths. *Concession, by deferred prosecution:* a $900 million forfeiture and an independent monitor, the charge deferred rather than a conviction entered. (U.S. DOJ, S.D.N.Y., 2015.)

2015. Peanut Corporation of America, Parnell. The company knowingly shipped salmonella-contaminated peanut products, tied to a 2008 to 2009 outbreak that killed nine. *Criminal:* Stewart Parnell was convicted on 67 counts and sentenced to 28 years, the first felony convictions of executives in a U.S. food-safety case. (*United States v. Parnell*, M.D. Ga., 2015.)

2015. Takeda, Actos. Bladder-cancer claims tied to the diabetes drug. *Settlement, without admission:* $2.4 billion. (Settlement, 2015.)

2015. Imperial Tobacco, Rothmans Benson and Hedges, and JTI-Macdonald, cigarettes, Quebec. The companies deceived the public and health authorities about the risks of smoking over decades. *Verdict:* the Quebec Superior Court found the three companies liable and awarded roughly CA$15.5 billion to two classes of Quebec smokers, upheld in its core by the Quebec Court of Appeal in 2019. (*Blais and Létourneau c. Imperial Tobacco*, Quebec Superior Court, 2015; affirmed 2019.)

2015. Royal Dutch Shell, Bodo, Nigeria. [Indigenous] Two 2008 and 2009 oil spills in the Ogoni

region of the Niger Delta. *Settlement:* £55 million out of court to the Bodo community. (Settlement, 2015.)

2016. BP, Deepwater Horizon, civil. Civil claims under the Clean Water Act and for natural-resource damage from the 2010 spill. *Settlement:* about $20.8 billion with the United States and five Gulf states, the largest single-entity settlement in Department of Justice history. (U.S. DOJ, 2015; entered 2016.)

2016. Wells Fargo. Millions of customer accounts and products were opened without authorization. *Regulatory:* $185 million to the Consumer Financial Protection Bureau, a federal bank regulator, and Los Angeles authorities. (CFPB and OCC, 2016.)

2017. Volkswagen AG, Dieselgate. Defeat-device software let roughly 590,000 U.S. diesel vehicles cheat emissions tests. *Criminal:* VW pleaded guilty to three felony counts and paid a $4.3 billion criminal and civil penalty; with prior civil settlements the U.S. total reached about $20 billion. (U.S. DOJ, E.D. Mich., 2017.)

2017. Takata Corporation. Airbag inflators could rupture and expel metal fragments. *Criminal:* Takata pleaded guilty to wire fraud and was sentenced to a $1 billion penalty, including restitution funds for victims and automakers. (U.S. DOJ, E.D. Mich., 2017.)

2017. DuPont and Chemours, C8 (PFOA), Ohio and West Virginia. The "forever chemical" used to make Teflon contaminated drinking water along the Ohio River. *Settlement, denying wrongdoing:* $670.7 million to resolve about 3,550 personal-injury claims. (Settlement, 2017.)

2017. Government of Canada, Sixties Scoop. [Indigenous] The removal of First Nations and Inuit children from their families into non-Indigenous homes. *Settlement:* an agreement of CA$800 million, most of it for individual compensation to survivors. (Government of Canada, 2017.)

2017. Ontario and Canada, Grassy Narrows mercury care. [Indigenous] The continuing burden of mercury poisoning on the Grassy Narrows people. *Concession:* Ontario legislated CA$85 million for river remediation and the federal government committed to fund a mercury care and treatment center. (Government commitments, 2017.)

2018. Johnson & Johnson, talc (Baby Powder). Plaintiffs alleged that talc contaminated with asbestos caused ovarian cancer; the company denies causation and denies asbestos content. *Verdict:* a Missouri jury awarded $4.69 billion to 22 women, reduced on appeal to about $2.12 billion, which upheld the core findings; the U.S. Supreme Court denied review in 2021. (Missouri courts, 2018 to 2020; certiorari denied 2021.)

2018. Monsanto, Roundup (glyphosate). A school groundskeeper, Dewayne Johnson, alleged that long-term Roundup exposure caused his non-Hodgkin lymphoma. *Verdict:* a jury awarded $289 million, reduced through appeals to more than $20.5 million, the liability finding upheld. (*Johnson v. Monsanto*, Cal., 2018.)

2018. Volkswagen AG, diesel defeat devices, Germany. The carmaker installed software in millions of diesel vehicles to cheat emissions tests and failed to prevent it. *Regulatory:* the Braunschweig public prosecutor imposed a 1 billion euro fine, which Volkswagen accepted without appeal. (Braunschweig Public Prosecutor's Office, 2018.)

2019. Insys Therapeutics and John Kapoor, Subsys fentanyl spray. The company bribed prescribers to write its fentanyl product, often for patients without cancer pain. *Criminal:* the founder and four executives were convicted of a RICO conspiracy, and the company paid $225 million before filing for bankruptcy. (D. Mass., 2019.)

2019. Sherwin-Williams, ConAgra Grocery Products, and NL Industries, lead paint, California. The companies knowingly marketed lead paint for interior residential use, leaving a lasting hazard in older homes.

Verdict and settlement: after a trial court found the products a public nuisance, affirmed in part on appeal in 2017, the three companies agreed in 2019 to pay a combined $305 million to ten California jurisdictions to abate lead-paint hazards in pre-1951 homes. (*People v. ConAgra Grocery Products*, California courts, 2017 to 2019.)

2019. Equifax, 2017 consumer-data breach. A failure to patch a known security flaw exposed the personal data of about 147 million people. *Regulatory settlement:* Equifax agreed to a global settlement with the Federal Trade Commission, the Consumer Financial Protection Bureau, and 48 states of at least $575 million and up to $700 million. (U.S. Federal Trade Commission, 2019.)

The 2020s

2020. Bayer and Monsanto, Roundup (glyphosate). Claims that long-term Roundup exposure caused non-Hodgkin lymphoma. *Settlement, without admission of liability:* about $10.9 billion to resolve roughly 125,000 claims, after three plaintiff verdicts survived appeal. (Bayer, 2020.)

2020. Monsanto, PCBs, Washington State. Statewide PCB contamination. *Settlement:* $95 million with the State of Washington. (Washington Attorney General, 2020.)

2020. Pacific Gas and Electric, 2018 Camp Fire. Utility equipment ignited a wildfire that killed 84 people. *Criminal:* PG&E pleaded guilty to 84 counts of involuntary manslaughter. (Butte County, 2020.)

2020. Blue Bell Creameries. Listeria-contaminated ice cream distributed in a 2015 outbreak was linked to three deaths. *Criminal:* the company pleaded guilty to two misdemeanor counts and paid $17.25 million in criminal penalties, with $2.1 million more in civil claims. (U.S. DOJ, W.D. Tex., 2020.)

2020. Wells Fargo. The unauthorized-accounts practices of 2002 to 2016. *Criminal and regulatory:* a $3 billion

resolution with the Department of Justice and the Securities and Exchange Commission, reached through a deferred prosecution agreement. (U.S. DOJ and SEC, 2020.)

2020. Purdue Pharma, OxyContin. [Indigenous among claimants] Defrauding federal regulators about efforts to curb overprescribing, and paying kickbacks. *Criminal:* Purdue pleaded guilty to three felony counts, with total criminal and civil exposure set at $8.3 billion; the wider opioid resolution included Native American tribal governments as settling parties. (U.S. DOJ, 2020.)

2021. Royal Dutch Shell, Niger Delta, Dutch court. [Indigenous] Pipeline spills that ruined farmland and fishing waters in the Niger Delta. *Verdict:* the Hague Court of Appeal found Shell's Nigerian subsidiary liable and recognized a parent-company duty of care. (Hague Court of Appeal, 2021.)

2021. Boeing, 737 MAX (MCAS flight-control system). After two crashes that killed 346 people, the company was charged over deceiving federal aviation regulators about the MCAS system. *Criminal, deferred prosecution:* Boeing resolved one count of conspiracy to defraud the United States and agreed to pay about $2.51 billion, comprising a $243.6 million criminal penalty, $500 million for the crash victims' heirs, and $1.77 billion to airline customers. (U.S. Department of Justice, N.D. Tex., 2021.)

2021. McKinsey and Company, opioid consulting. The firm advised opioid makers, including Purdue Pharma, on tactics to boost sales, such as targeting high-volume prescribers. *Settlement, without admission of wrongdoing:* $573 million to 47 states and other jurisdictions, with a release of internal documents and adopted restrictions. (Multistate attorneys general, 2021.)

2022. McKesson, Cardinal Health, AmerisourceBergen, and Johnson & Johnson, opioid distribution and marketing. The three largest

U.S. drug distributors and a manufacturer were alleged to have fueled the opioid epidemic through their distribution and promotion. *Settlement, without admission of liability:* about $26 billion over up to eighteen years, the distributors paying roughly $21 billion and Johnson & Johnson up to $5 billion, with a mandated independent order-monitoring clearinghouse. (National opioids settlement, agreed 2021, effective 2022.)

2022. JUUL Labs, e-cigarettes. The company marketed its nicotine vaping products in ways that reached and appealed to underage users. *Settlement:* a $438.5 million multistate settlement with 34 states and territories, with sweeping restrictions on its marketing, sales, and distribution. (Multistate attorneys general, 2022.)

2023. 3M, PFAS in public drinking water. "Forever chemicals" detected in public water systems across the country. *Settlement, without admission of liability:* up to $10.3 billion in present value, roughly $12.5 billion over thirteen years, to public water suppliers. (3M, 2023; court approval 2024.)

2023. DuPont, Chemours, and Corteva, PFAS. PFAS contamination of public water systems. *Settlement, without admission:* $1.185 billion. (Settlement, 2023.)

2023. 3M, Combat Arms earplugs. Service members alleged hearing loss and tinnitus from earplugs that fit poorly. *Settlement, without admission of liability:* $6.01 billion to resolve some 250,000 claims. (*In re 3M Combat Arms Earplug*, N.D. Fla., 2023.)

2024. BHP, Vale, and Samarco, Mariana dam collapse, Brazil. [Indigenous among affected] The 2015 collapse of the Fundão tailings dam sent mine waste down the Rio Doce, harming downstream communities including the Krenak. *Settlement:* a reparation agreement with Brazilian authorities valued at R$170 billion, roughly US$31.7 billion, in past and future obligations over some twenty years. (Definitive settlement, ratified 2024.)

What the list is for

Taken in order, the years reveal a pattern that no single case could. It is not any single case. Any one company can be called a bad actor, any one settlement an aberration. It is the length of the list, and the sameness of it: the same conduct, in the same forms, decade after decade, in medicine and chemicals and cars and food and fuel and paper and finance, across four continents, with no sign of thinning as it nears the present. This book has argued that which plants and products a society punishes, and which it sells, is decided by money and power more than by danger. This is the evidence for that argument, taken from a different room than the drug laws: the civil and criminal courts, where the harms profitable enough to be worth defending are finally written down. What the record shows is a system that harms and pays, and harms and pays again, and has generally found it cheaper to continue than to stop. Whether that outcome is the system's purpose is not asserted here. The purpose of a system, it has been said, is what it does. The reader has the record, and the conclusion is the reader's to draw.

These are the cases where someone was made to answer. The chapters that close the account turn to the part of the bill that is not settled in any court: who pays for all of it in their bodies, and who is still paying.

PART FOUR

The Reckoning



Chapter Twelve

Who Pays



On a spring morning two farmers walk out to their fields. One is in the Bolivian Andes, where his family has grown coca longer than anyone can count, and he will tuck a few leaves into his cheek against the thin air and the long day, the way his people did before the Inca. The other is in Kentucky, where his family has grown tobacco for generations, and his crop will be cured, blended, and sold in packs that kill on the order of half a million Americans a year. By the letter of the law the first man is a criminal and his plant is contraband the nations of the world have sworn to abolish, while the second man's crop is licensed, taxed, and, for most of the last century, advertised. Both raise a plant that can harm. Only one of them is treated as a danger. This chapter asks what actually decided which.

The answer is a pattern, and it must be called a pattern and not a plan, because the record shows not a single hidden hand but a consistent asymmetry, repeated across a century of statutes and court rulings and enforcement data. When a plant is bound in the public mind to a people outside the seat of power, the poor, the colonized, the racialized, the response is prohibition, and the punishment falls on that people. When a substance is profitable and belongs to the majority or the powerful, the response is regulation, taxation, and tolerance, even when the profitable substance kills far more. That asymmetry is not asserted here as an opinion. It is on the record, as the rest of this chapter sets out to illustrate.

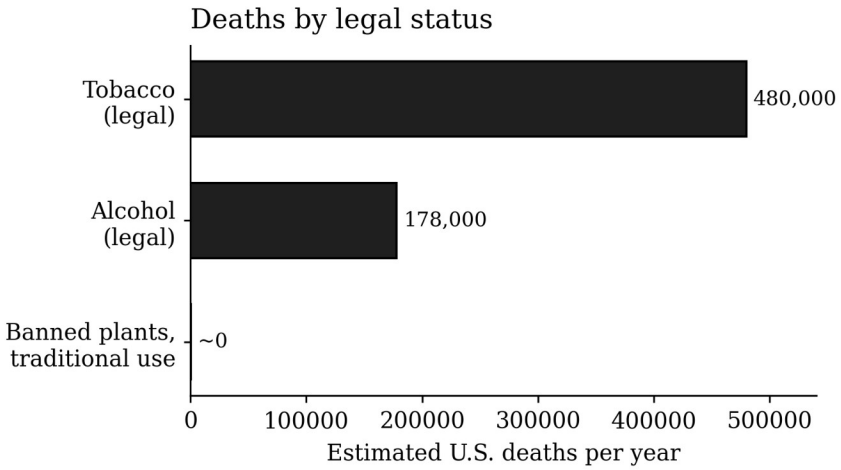


"The Anti-Chinese Wall," Puck, 1882.

The first drug law in the country already set the template every later ban would follow: a prohibition aimed at a people, not at a danger. San Francisco's 1875 ordinance against opium dens, generally counted the first anti-drug law in America, targeted the Chinese method of taking a drug that white households were consuming freely in other forms, and the fear it named aloud was of white men and women being corrupted in Chinese establishments. It sat inside the same years that produced the Chinese Exclusion Act of 1882. Then came the plant renamed: cannabis, long known as Indian hemp, was recast as the foreign-sounding marihuana, and Harry Anslinger's campaign for the 1937 federal ban ran on open racial fear, tying the drug to Mexican and Black Americans. The motive behind the statute was mixed and is still debated, but the modern measure of who paid is not: the American Civil Liberties Union found that Black Americans were arrested for cannabis three and a half to nearly four times as often as white Americans, despite using it at the same rate, some eight million arrests in a single decade, nearly nine in ten for simple possession. Then the coca leaf, swept into Schedule I of the 1961 Single Convention, whose Article 49 ordered coca chewing abolished within twenty-five

years, so that Bolivia had to quit the treaty and rejoin it with a reservation to keep a practice thousands of years old. Then peyote, whose protection the Supreme Court removed in 1990 and which took an act of Congress to restore in 1994, a right the majority never had to litigate for its own intoxicants. Then crack cocaine, which the 1986 law punished a hundred times as heavily as the pharmacologically near-identical powder, so that by the government's own count around four in five federal crack offenders were Black. The imprisonment gap for drug offenses ran as high as fifteen to one at the turn of the century and had narrowed to roughly five to one by 2016, smaller than it was, and still large.

All of that sits against the ledger of harm, and against what these chapters have said before: the purpose of the system is what the system does. This system works to extract wealth, whatever it costs a population's health, and to punish the people who choose to consume a plant no authority has approved. Tobacco kills about four hundred and eighty thousand Americans a year and alcohol about a hundred and seventy-eight thousand, and both were legal, taxed, and advertised the entire time. The plants prohibited earliest and hardest carry, in their traditional forms, death tolls near zero. And the deadliest drug epidemic of the age came not from any banned plant but from an approved, marketed pharmaceutical, OxyContin, a product a federal court found had been sold by fraud. The line between the criminal and the customer was never drawn by danger. It was drawn by who profited from the sale and who was doing the consuming.



Estimated U.S. deaths per year, by legal status.

It is not only an American habit. Through the eighteenth and nineteenth centuries the British East India Company grew opium in India and drove it into China, from a few hundred chests a year to some forty thousand, and when the Chinese government tried to enforce its own prohibition, Britain fought two wars to keep the trade open and forced China, at gunpoint, to legalize it. The same imperial power that would later help write the world's drug-control treaties once waged war to defeat a drug prohibition, because that drug was profitable to its ruling class. Colonial governments ran opium as a licensed state monopoly, collecting the revenue while punishing unlicensed use among their subjects. The actors change from empire to empire and century to century. The incentive does not.

The pattern has limits, and those limits matter. Cannabis and the opium poppy, two of the most heavily prohibited plants, are not New World or Indigenous plants at all; their origins lie in Asia. What was racialized was not the plant itself but the people associated with its use, and the laws imposed upon them. The motives behind any individual law were mixed, and no honest reading can reduce an entire statute to a single intention. The enforcement gaps, cruel as they were, have narrowed. None of that erases the larger

pattern. Across too many countries and too many decades, the plants associated with powerless people were more likely to be prohibited and their users punished, while substances tied to wealth and political power were more likely to be licensed, taxed, and defended, sometimes even by war. The pattern does not require a single conspiracy or a single motive. It requires only an asymmetry in whose interests the law protected. Who paid was decided, again and again, by who held power. And the same arithmetic that decided who went to prison also decided whose bodies would carry the sickness of the world the powerful built. That is the last question this book has to answer.

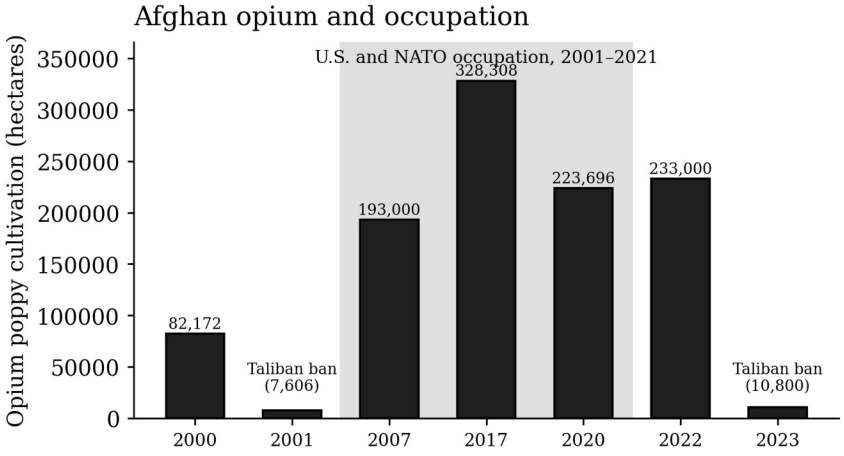
Chapter Thirteen

The Disease and the Cure



In the spring of 1925 a small group of men gathered in Washington to decide whether the country could keep selling leaded gasoline. The Surgeon General had called them together after the deaths at the refineries, and among them sat the representatives of the companies that made the additive, General Motors, DuPont, and Standard Oil. The committee did not so much approve the poison as decline to forbid it, and it asked for further study. But the study was never funded by the government. It was left, in effect, to the industry's own laboratory, which spent the next forty years finding the lead safe. The episode offers a simple example of the problem this chapter examines: the institution responsible for protecting the public allowed the industry whose product was under question to become one of the principal authorities determining whether that product was dangerous.

The name for this is regulatory capture, and it is not a slur but a settled idea in mainstream economics. The economist George Stigler, who won a Nobel Prize in part for the work, put it plainly in 1971: as a rule, he wrote, regulation is acquired by the industry and is designed and operated primarily for its benefit. His reasoning was simple. The industry being regulated is concentrated, wealthy, and organized, and it cares enormously about the rules; the public is dispersed and distracted and pays little attention. The concentrated side reliably wins. Capture is not a conspiracy. It is what the incentives predict and what the system allows.



Afghan opium-poppy cultivation and the years of the U.S. and NATO occupation. Source: UN Office on Drugs and Crime.

It is visible in the government now. The Food and Drug Administration, which decides what drugs Americans may take, is today funded in large part by fees paid by the drug companies themselves; those industry fees cover roughly two-thirds of its drug-review budget. The federal committee that tells Americans what to eat was found, in a study of its 2020 membership, to carry more than seven hundred financial ties to the food and pharmaceutical industries. And the same people move back and forth across the line, in what is called the revolving door. One official, Michael Taylor, passed between the FDA, a law firm that represented the agricultural giant Monsanto, the Department of Agriculture, and Monsanto itself. None of this proves that any single decision was bought. It is a documented structural conflict, the very one Stigler's theory predicts, and careful scholars caution that capture is real but neither total nor inevitable. Still the shape is undeniable: the referee is now substantially paid by the players.

The profession that was meant to be the last honest broker has, at documented moments, been an arm of the sale, and the shift is visible even in ordinary practice. House calls were once standard; a doctor took the time to learn a patient's history and to weigh the several symptoms as one puzzle to

be solved. Today a great deal of care is structured differently. The doctor asks questions from behind a computer terminal, and as the patient lists symptoms the answers are entered and a prescription comes back for the disease the software names. Each symptom is treated on its own, and each with an approved, patented drug; if there are several symptoms there are several drugs, while the underlying cause goes unexplored unless the patient knows enough to press for it. The body is seldom viewed as a whole. One might say that at least the physician no longer sells from the page as he once did, but the result can be as bad as the cure. In the 1940s and 1950s the tobacco companies put physicians in their advertisements, More Doctors Smoke Camels, and the medical journals took the money; the Journal of the American Medical Association ran cigarette advertising until 1953. Half a century later the same authority was rented again. Pain was declared the fifth vital sign, a clinicians' idea that the opioid makers amplified and funded, written into hospital accreditation standards by 2001 and taught through more than twenty thousand industry-sponsored pain courses that urged doctors to prescribe; Purdue's own records treated the phrase as a marketing campaign. The authority of medicine was rented out twice, a century apart, to move the two deadliest legal drugs in the country.



An amputation before anaesthesia, 1793 (after Thomas Rowlandson).

The motif these chapters have named throughout reaches its sharpest form here. In case after case the same corporate structure stands on both sides of a harm, profiting when people are sick and again when they try to get well. Sometimes this is only a company following the market it helped make, as when the food conglomerates that engineered the processed diet began selling meals for users of the new weight-loss drugs. Sometimes the same firms sell both an addiction and the product marketed to quit it. And once it reached the grotesque, when a member of the Sackler family, whose company did more than any other to drive the opioid epidemic, was granted a patent on a treatment for opioid addiction, though the company said no profit would be taken from it. What the documents prove is the adjacency and the incentive, recurring across food and tobacco and opioids. What they do not prove, and what these pages do not claim, is a single coordinated plan to sicken people in order to sell them the cure. The pattern of the system is damning enough to determine the purpose.

From a step back, the separate scandals resolve into a single system. The plants that paid were abundantly available and promoted: tobacco and alcohol, the deadliest of all, advertised and taxed for a century. The plants that did not pay were whimsically or ideologically forbidden and their keepers punished: coca and peyote, the mushroom and the vine, cannabis and the poppy. The food was conscripted, tuned to the edge of craving. The referee was captured, the profession was rented, and the science was bought. These are not a list of separate wrongs. They are a single environment, built at every level to profit from what it does to the human body, one in which the profitable harm is legal and defended, the unprofitable remedy is forbidden or buried, and the bill comes due in other people's bodies.



Cocaine Toothache Drops, Lloyd Manufacturing Co., 1885: cocaine sold over the counter for children's toothache, at fifteen cents the bottle. The remedy as dangerous as the disease.

It comes due hardest on the people this series began with. The nations of the first two books, who lived in a worked-out balance with the plants and the land, now carry the highest rates of diabetes, of alcohol death, and of drug overdose of any people in the country. The cause is not in their blood, and the proof is on the record. The Pima of Arizona have among the highest rates of type-2 diabetes ever measured,

close to two in five adults; the Pima of Mexico, the same people by ancestry, living on the old farming diet, have about a sixth as much. The same genome, and six times the disease: the difference is the world around it, not the blood within it. And the change to that world was done to them. The river that watered the Arizona Pima's fields was dammed and drawn off upstream, their farming ruined, and the harvest replaced by government rations of lard and white flour and sugar. The old story that Native people carry a thrifty gene that dooms them to diabetes has been laid beside that dry riverbed, and it does not survive the comparison. The reciprocal world of the first two books was not merely lost. It was replaced by an environment engineered to sicken the very people it was built on top of.

This is a pattern, not a proven plot; an adjacency of incentives, not a secret scheme. The sickness is structural, not biological. Capture is real but not total, and the record holds the counter-movements too: the tribes reclaiming their old foods, the settlements that at last named the harm, the slow return of the forbidden sacraments to the people who kept them. None of it is fate. And then, at the end, return to where the whole series began, to a pinch of tobacco laid in the ground before a root was taken, offered by a people who believed they owed the land a debt for all they drew from it. That plant became the deadliest product the country ever sold. Every plant in these pages was, to someone, once a gift, a food or a medicine or a road to the sacred, and each was turned, by money and power, into either a weapon or a crime. A civilization can be measured by what it does with such gifts. This one decided which plants to forbid and which to sell by asking not what they do to us, but who profits from the answer. The people who first offered the tobacco are still paying for that decision, and what is owed to them is the last thing these pages have to say.


VIN MARIANI

Appendix: A Fuller Record, by Year and Name

The cases told above are only the ones set down in full. What follows is a longer reference list of further documented cases, name and year and outcome type alone, in which a company or a government was held to account for harm to people or the living world. It does not repeat the cases above, and it is itself only a portion of the record. Cases marked [Indigenous] concern harm to an Indigenous people or nation.

The 1950s and 1960s

- 1967. Torrey Canyon, United Kingdom and France, tanker oil pollution. Settlement.
- 1967. Aberfan, Wales, National Coal Board, colliery-tip collapse. Regulatory.

The 1970s

- 1974. Reserve Mining, asbestos fibers in Lake Superior. Verdict.
- 1976. Seveso, Italy, Icmesa and Givaudan, dioxin release. Concession.
- 1979. Silkwood v. Kerr-McGee, plutonium contamination. Verdict.

The 1980s

- 1988. Johns-Manville asbestos trust, asbestos disease. Settlement.

The 1990s

- 1997. Mississippi tobacco settlement, smoking illness and health costs. Settlement.
- 1997. Florida tobacco settlement, smoking illness and health costs. Settlement.
- 1998. Texas tobacco settlement, smoking illness and health costs. Settlement.

- 1998. Minnesota tobacco settlement, smoking illness and health costs. Settlement.
- 1998. Ngāi Tahu Deed of Settlement, New Zealand, land and resource loss. Concession. [Indigenous]
- 1998. Diesel engine makers, Clean Air Act, seven manufacturers. Settlement.

The 2000s

- 2000. Koch Petroleum, refinery air pollution. Settlement.
- 2001. BP, refinery air pollution. Settlement.
- 2001. Marathon Ashland Petroleum, refinery air pollution. Settlement.
- 2001. Motiva and Shell Deer Park, refinery air pollution. Settlement.
- 2002. Schering-Plough, Claritin, manufacturing violations. Settlement.
- 2003. Chevron, refinery air pollution. Settlement.
- 2003. Colonial Pipeline, Clean Water Act, oil spills. Settlement.
- 2003. AstraZeneca, Zoladex, Medicare fraud. Settlement.
- 2004. CITGO, refinery air pollution. Settlement.
- 2004. Plan de Sánchez v. Guatemala, massacre and state responsibility. Verdict. [Indigenous]
- 2005. Valero, refinery air pollution. Settlement.
- 2005. Sunoco, refinery air pollution. Settlement.
- 2005. ConocoPhillips, refinery air pollution. Settlement.
- 2005. Serono, Serostim, off-label promotion and kickbacks. Settlement.
- 2005. Yakye Axa Community v. Paraguay, land dispossession and deaths. Verdict. [Indigenous]

- 2005. Moiwana Community v. Suriname, massacre and displacement. Verdict. [Indigenous]
- 2006. Sawhoyamaxa Community v. Paraguay, land dispossession and deaths. Verdict. [Indigenous]
- 2006. Schering-Plough, Temodar and Intron A, off-label promotion and kickbacks. Settlement.
- 2006. Owens Corning asbestos trust, asbestos disease. Settlement.
- 2007. Bristol-Myers Squibb, Abilify and Serzone, off-label promotion and kickbacks. Settlement.
- 2007. Saramaka People v. Suriname, logging concessions and land. Verdict. [Indigenous]
- 2008. Cephalon, Actiq and Provigil, off-label promotion. Settlement.
- 2008. Massey Energy, Clean Water Act, coal-mine wastewater. Settlement.

The 2010s

- 2010. Allergan, Botox, off-label promotion. Settlement.
- 2010. Novartis, Trileptal, off-label promotion and kickbacks. Settlement.
- 2010. Xákmok Kásek Community v. Paraguay, land dispossession and deaths. Verdict. [Indigenous]
- 2010. Guidant and Boston Scientific, defective defibrillators. Criminal.
- 2011. Medtronic, device kickbacks. Settlement.
- 2012. Boehringer Ingelheim, Aggrenox and Combivent, off-label promotion and kickbacks. Settlement.
- 2012. Barclays, LIBOR rate rigging. Regulatory.
- 2012. UBS, LIBOR rate rigging. Regulatory.
- 2013. JPMorgan Chase, mortgage-backed securities. Settlement.

- 2013. Wyeth, Rapamune, off-label promotion. Criminal.
- 2013. Royal Bank of Scotland, LIBOR rate rigging. Regulatory.
- 2013. Tūhoe Deed of Settlement, New Zealand, confiscation in the Urewera. Concession. [Indigenous]
- 2014. Bank of America, mortgage-backed securities. Settlement.
- 2014. Citigroup, mortgage-backed securities. Settlement.
- 2014. Dole, Nicaragua banana workers, DBCP pesticide injury. Settlement.
- 2014. W.R. Grace, Superfund cost recovery, thirty-nine contaminated sites. Settlement.
- 2014. Ngāruahine Deed of Settlement, New Zealand, land loss in Taranaki. Concession. [Indigenous]
- 2015. Rangitāne o Manawatū Deed of Settlement, New Zealand, land loss. Concession. [Indigenous]
- 2015. Kaliña and Lokono Peoples v. Suriname, land and a nature reserve. Verdict. [Indigenous]
- 2015. Garífuna Triunfo de la Cruz Community v. Honduras, land and development. Verdict. [Indigenous]
- 2015. Garífuna Punta Piedra Community v. Honduras, land dispossession. Verdict. [Indigenous]
- 2015. Deutsche Bank, LIBOR rate rigging. Regulatory.
- 2015. Global banks, foreign-exchange rigging, guilty pleas. Criminal.
- 2015. Warner Chilcott, health-care fraud and kickbacks. Criminal.
- 2016. Goldman Sachs, mortgage-backed securities. Settlement.
- 2016. Genentech and OSI, Tarceva, off-label promotion and misleading data. Settlement.

- 2016. Hyundai and Kia, fuel-economy misrepresentation. Settlement.
- 2016. Enbridge, Clean Water Act, Kalamazoo and Illinois oil spills. Settlement.
- 2016. Chichupac and Rabinal communities v. Guatemala, massacres and displacement. Verdict. [Indigenous]
- 2016. Pacific Gas and Electric, San Bruno pipeline, fatal gas explosion. Criminal.
- 2016. Salix Pharmaceuticals, kickbacks through speaker programs. Settlement.
- 2016. Mylan, EpiPen, Medicaid rebate misclassification. Settlement.
- 2017. AmerisourceBergen, adulterated repackaged drugs and kickbacks. Settlement.
- 2017. Shire, device and drug kickbacks. Settlement.
- 2017. Aegerion, Juxtapid, off-label promotion. Criminal.
- 2017. Celgene, Thalomid and Revlimid, off-label promotion and kickbacks. Settlement.
- 2017. Home Depot, payment-card data breach. Settlement.
- 2017. Anthem, health-data breach. Settlement.
- 2018. Xucuru People v. Brazil, land titling and dispossession. Verdict. [Indigenous]
- 2019. Fiat Chrysler, diesel emissions defeat devices. Settlement.
- 2019. Reckitt Benckiser and Indivior, Suboxone, opioid marketing fraud. Criminal.
- 2019. Johnson & Johnson, Oklahoma opioid trial, public nuisance. Verdict.
- 2019. Facebook, privacy violations. Regulatory.
- 2019. Federal Indian Day Schools, Canada, the McLean class action. Settlement. [Indigenous]

The 2020s

- 2020. Daimler and Mercedes-Benz, diesel emissions defeat devices. Settlement.
- 2020. Novartis, kickbacks through sham speaker programs. Settlement.
- 2020. Mallinckrodt, Acthar Gel, Medicaid underpayment and kickbacks. Settlement.
- 2020. Actelion, pulmonary-hypertension drugs, co-pay kickbacks. Settlement.
- 2020. Gilead Sciences, HIV drugs, kickbacks. Settlement.
- 2020. Sanofi, co-pay foundation kickbacks. Settlement.
- 2020. Hyundai and Kia, delayed engine-fire recalls. Regulatory.
- 2020. British Airways, data breach. Regulatory.
- 2020. Marriott International, data breach. Regulatory.
- 2020. Lhaka Honhat Association v. Argentina, land, environment, and food. Verdict. [Indigenous]
- 2021. Vale, Brumadinho, Brazil, tailings-dam collapse, 270 deaths. Settlement.
- 2021. Capital One, data breach of 100 million records. Settlement.
- 2021. Atlantic Richfield, Anaconda Smelter, Montana, Superfund contamination. Settlement.
- 2022. CVS, Walgreens, and Walmart, opioid dispensing. Settlement.
- 2022. Teva and Allergan, opioid manufacturing and marketing. Settlement.
- 2022. African Court, Ogiek v. Kenya, Mau Forest evictions and reparations. Verdict. [Indigenous]
- 2022. T-Mobile, data breach of 77 million records. Settlement.

- 2022. First Nations drinking water, Canada, long-term unsafe-water advisories. Settlement. [Indigenous]
- 2022. Mallinckrodt, opioid bankruptcy resolution. Settlement.
- 2023. Robinson Huron Treaty annuities, Canada, unpaid treaty annuities. Settlement. [Indigenous]
- 2025. Omnicare and CVS, fraudulent dispensing and false claims. Verdict.
- 2025. Mercedes-Benz and Daimler, multistate diesel-emissions fraud. Settlement.

A note on scale

- 2011 onward. United Kingdom banks, payment-protection-insurance mis-selling, consumer redress in the tens of billions of pounds. Regulatory.

This list is not exhaustive. It omits, among much else, the greater part of a hundred or more Department of Justice pharmaceutical and medical-device fraud settlements, the scores of asbestos bankruptcy trusts, the many Clean Air and Clean Water Act consent decrees, and the seventy and more New Zealand Treaty of Waitangi settlements, each of which carries a documented outcome of its own.

Appendix: Schedule I

SCHEDULE I OF THE CONTROLLED SUBSTANCES ACT

A drug or substance is placed in Schedule I on three findings: (A) it has a high potential for abuse; (B) it has no currently accepted medical use in treatment in the United States; and (C) there is a lack of accepted safety for its use under medical supervision.

Among the substances so classified: heroin; cannabis (marihuana); peyote; mescaline; psilocybin and psilocyn; dimethyltryptamine (DMT); MDMA; and LSD.

Every plant treated in this book, except tobacco and alcohol, appears in this tier or had to be exempted from it by a separate act of Congress or a court.

21 U.S.C. § 812.

A Note on Sources

This is an argument, and an argument is only as good as what it is built on. The claims in these chapters rest, wherever possible, on the kind of evidence that does not depend on trusting the author: the findings of courts, the reports of government bodies, the studies in the peer-reviewed literature, and the internal records that litigation forced into public view, in which companies set down in their own words what they knew. Where a claim rests on something weaker, the text says so. Where the record establishes only that two things happened side by side, it is called an adjacency and not a scheme. The distinction between a documented pattern and an alleged plan is kept throughout, on purpose, because the pattern is strong enough on its own and does not need to be inflated.

What follows is not an exhaustive citation apparatus but a guide to the principal works and documents behind each part, for readers who want to go to the record themselves.

The primary record

Several primary documents run through the whole book. The most important is the ruling in *United States v. Philip Morris USA* (2006), in which Judge Gladys Kessler of the federal district court in Washington found, after a lengthy trial, that the major cigarette manufacturers had carried on a decades-long scheme to defraud the public about the harms of smoking. The tobacco companies' own memoranda, including the 1969 Brown and Williamson document containing the line "doubt is our product," are preserved in the Truth Tobacco Industry Documents archive maintained by the University of California, San Francisco. On the drug laws, the foundational instruments are the Harrison Narcotics Act of 1914, the Marihuana Tax Act of 1937, the Controlled Substances Act of 1970, and, internationally, the 1961 Single Convention on Narcotic Drugs, whose Article 49 ordered the abolition of coca chewing. The peyote and ayahuasca cases turn on *Employment Division v. Smith*

(1990), the Religious Freedom Restoration Act of 1993, the 1994 amendment to the American Indian Religious Freedom Act, and *Gonzales v. O Centro Espírita Beneficente União do Vegetal* (2006). On the forever chemicals, the findings of the court-mandated C8 Science Panel stand behind the account in the playbook chapter.

Part One: Tobacco and Corn

On tobacco, the essential histories are Allan Brandt's *The Cigarette Century* and Robert Proctor's *Golden Holocaust*; on the racial targeting of menthol, Keith Wailoo's *Pushing Cool*. The sacred and reciprocal use of tobacco and corn that opens these chapters draws on the record assembled in the first two books of this series. On alcohol and the fur trade, Peter Mancall's *Deadly Medicine* documents the trade in liquor with Indigenous nations; the debunking of the "firewater" metabolism myth follows the medical and historical literature cited in the chapter.

Part Two: The Forbidden Plants

On cannabis, Isaac Campos's *Home Grown* and Bonnie and Whitebread's *The Marihuana Conviction* trace the racialized origins of prohibition; the enforcement figures are from the American Civil Liberties Union's reports on marijuana arrests. On coca, the indispensable history is Paul Gootenberg's *Andean Cocaine*. On peyote, Thomas Maroukis's *The Peyote Road* and the litigation record above. On the opium poppy and the modern opioid crisis, David Courtwright's *Dark Paradise*, Patrick Radden Keefe's *Empire of Pain*, and Beth Macy's *Dopesick*, together with the Purdue Pharma bankruptcy record and *Harrington v. Purdue Pharma* (2024). The consciousness chapters on ayahuasca and psilocybin rest for their documented spine on the church-suppression and legal record; the broader and more speculative claims raised in them, discussed below, are treated in the text as open questions rather than established fact.

Part Three: The Playbook

The method chapter draws on Gerald Markowitz and David Rosner's work on lead and industrial poisons, Naomi Oreskes and Erik Conway's *Merchants of Doubt*, and David Michaels's *Doubt Is Their Product* and *The Triumph of Doubt*. On the migration of that method into food, the central works are Michael Moss's *Salt Sugar Fat* and *Hooked* and David Kessler's *The End of Overeating*, with the empirical link between tobacco ownership and hyperpalatable formulation drawn from the published research of Tera Fazzino and colleagues, and the addiction comparison from Ashley Gearhardt and colleagues.

Part Four: The Reckoning

On the racial history of drug enforcement, James Forman Jr.'s *Locking Up Our Own* and Michelle Alexander's *The New Jim Crow* are read alongside one another, as the text notes, because they disagree in useful ways. On regulatory capture, the foundational statement is George Stigler's 1971 paper "The Theory of Economic Regulation." On the Opium Wars and the imperial drug trade, the standard diplomatic and economic histories. The Arizona-and-Mexico Pima comparison, the natural experiment that closes the argument on Indigenous health, is drawn from the published epidemiological literature on the two populations.

On the contested works

Three of the chapters, particularly those on the visionary plants and the closing argument, touch questions that lie beyond the documented record: what the psychedelic experience is, where it came from, and how far the institutions that govern medicine and drugs can be trusted. A number of writers who press hard on those questions were read in the course of this book, among them Brian Muraresku, Rick Strassman, Andrew Gallimore, and Graham Hancock. Their more sweeping claims are not treated here as proven, and the reader will find them flagged in the text as hypotheses rather than facts. They are named because the

questions they raise are real and because honesty requires acknowledging where the settled record ends and the argument passes into open country. The load-bearing claims of this book do not rest on them. They rest on the documents.

About The Forgotten Series

The Forbidden Plants is the third book of The Forgotten Series. The first, *The Forgotten Harvest*, is about the plants themselves and the people who knew them. The second, *The Forgotten Wisdom*, is about how those people lived with the plants and the land, inside a bargain that returned what it took. This third book is about what was done to that world, and to the plants that once held it together. The three are meant to be read as one long account: a gift, a way of living by it, and the undoing of both.